1.4.7. Amplifier Saturation

- **limited linear range** – practically, amplifier operation is linear over a limited input range.
- **saturation** – beyond this range, saturation occurs.
  - output remains constant as input varies

$$\underbrace{\frac{L_{minus}}{A_v} \le v_i \le \frac{L_{plus}}{A_v}}_{\text{or...}\quad L_{minus} \le v_o \le L_{plus}}$$

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2.4.2. The Instrumentation Amplifier

- **Q:** What is one problem associated with the difference amplifier?
  - **A:** Low input impedance.
- **Q:** And, what does this mean practically?
  - **A:** That source impedance will have an effect on gain.
- **Q:** What is the solution?
  - **A:** Placement of two buffers at the input terminals, amplifiers which transmit the voltage level but draw minimal current.

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A Shift in Notation

- Before this point…
  - The parameter *A* is used to represent open-loop gain of an op amp.
  - The parameter *G* is used to represent ideal / non-ideal closed-loop gain of an op amp.
- After this point…
  - The parameter *A* is used to represent ideal gain of an op amp in a given closed-loop configuration.
  - The parameter *G* is not used.

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2.4.1. A Single Op-Amp Difference Amp

- **Q:** What are the characteristics of the difference amplifier?
  - **A:** Refer to following equations…

$$v_{Out} = \frac{(R_2 + R_1)R_4}{(R_4 + R_3)R_1} v_{In2} - \frac{R_2}{R_1} v_{In1}$$

but if $\begin{Bmatrix} R_1 = R_3 \\ R_2 = R_4 \end{Bmatrix}$ then $v_{Out} = \frac{R_2}{R_1}\left(v_{In2} - v_{In1}\right)$

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2.4. Difference Amplifiers

- **Q:** The op amp itself is differential in nature, why cannot it be used by itself?
  - **A:** It has an infinite gain, and therefore cannot be used by itself. One must devise a closed-loop configuration which facilitates this operation.

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2.4. Difference Amplifiers

- **common-mode rejection ratio** (CMRR) – is the degree to which a differential amplifier "rejects" the common-mode input.
  - Ideally, CMRR = *infinity*…

$$CMRR = 20 \log_{10} \left| \frac{A}{A_{Cm}} \right|$$

2.4. Difference Amplifiers

common-mode input

common-mode gain

differential input

differential gain

$$v_{Out} = A v_{dfi} + A_{cm} v_{cmi}$$

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2.4. Difference Amplifiers

- **difference amplifier** – is a closed-loop configuration which responds to the difference between two signals applied at its input and ideally rejects signals that are common to the two.
  - Ideally, the amp will amplify only the differential signal (v_{dfi}) and reject completely the common-mode input signal (v_{cmi}). However, a practical circuit will behave as below…

$$v_{Out} = Av_{dfi} + A_{cm}v_{cmi}$$

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Configuration and Characteristics of Buffer / Voltage-Follower Op-Amp Configuration

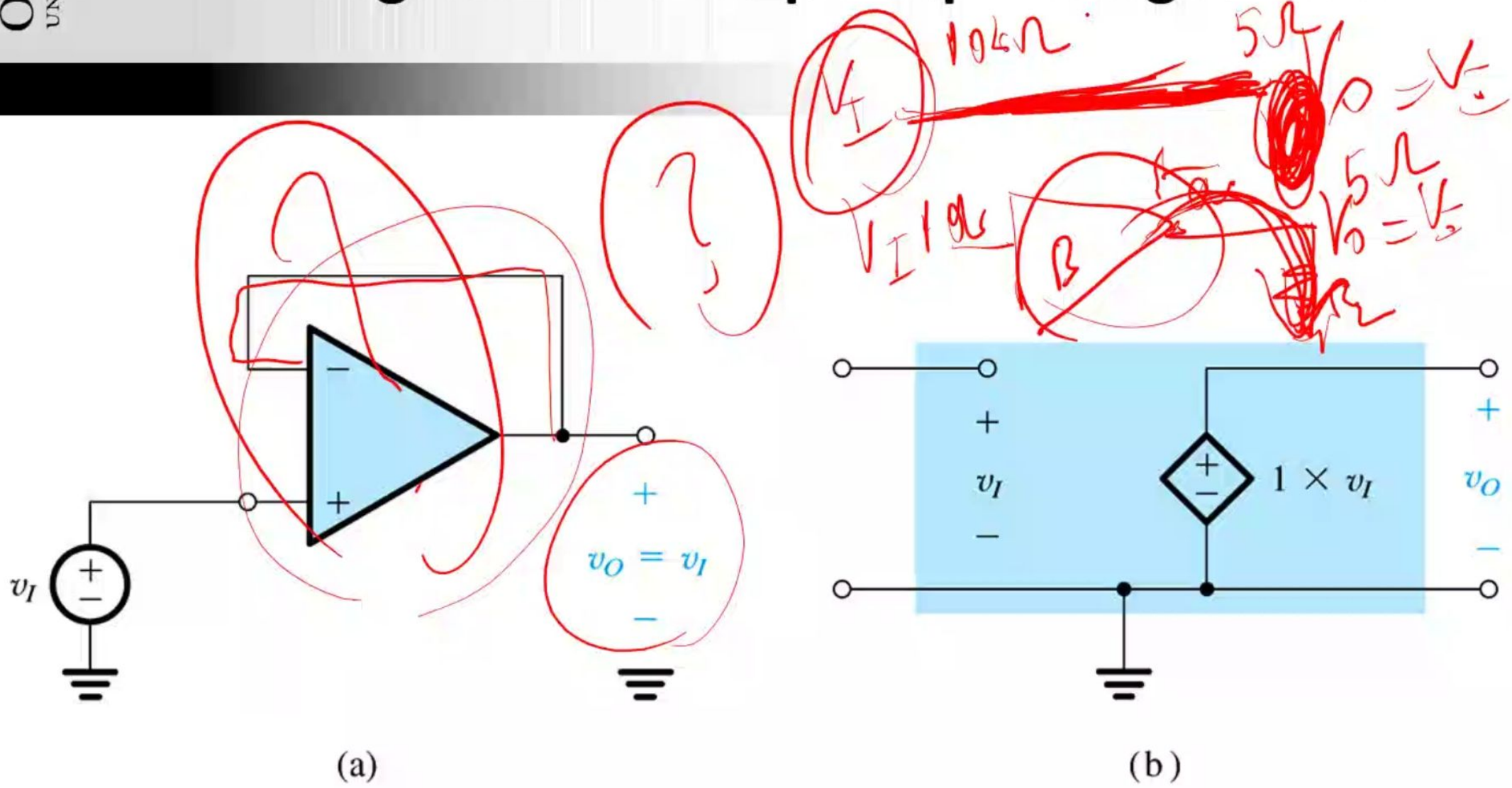


Figure 2.14: (**a**) The unity-gain buffer or follower amplifier. (**b**) Its equivalent circuit model.

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Configuration and Characteristics of Buffer / Voltage-Follower Op-Amp Configuration

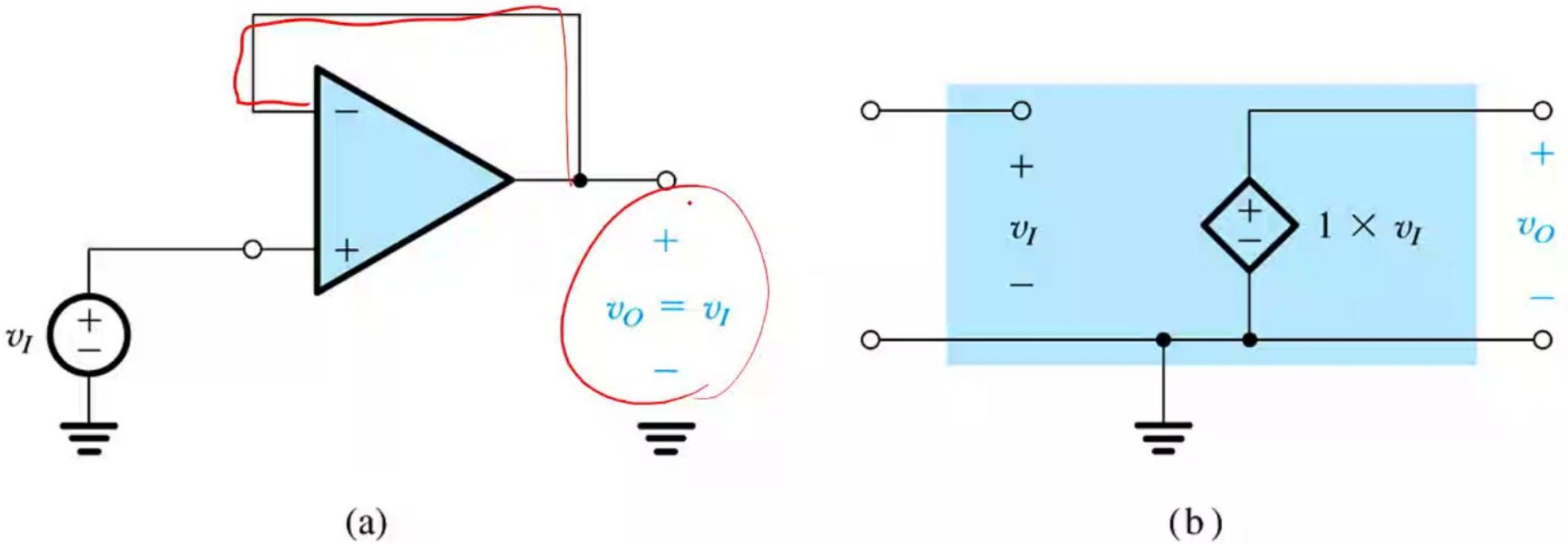
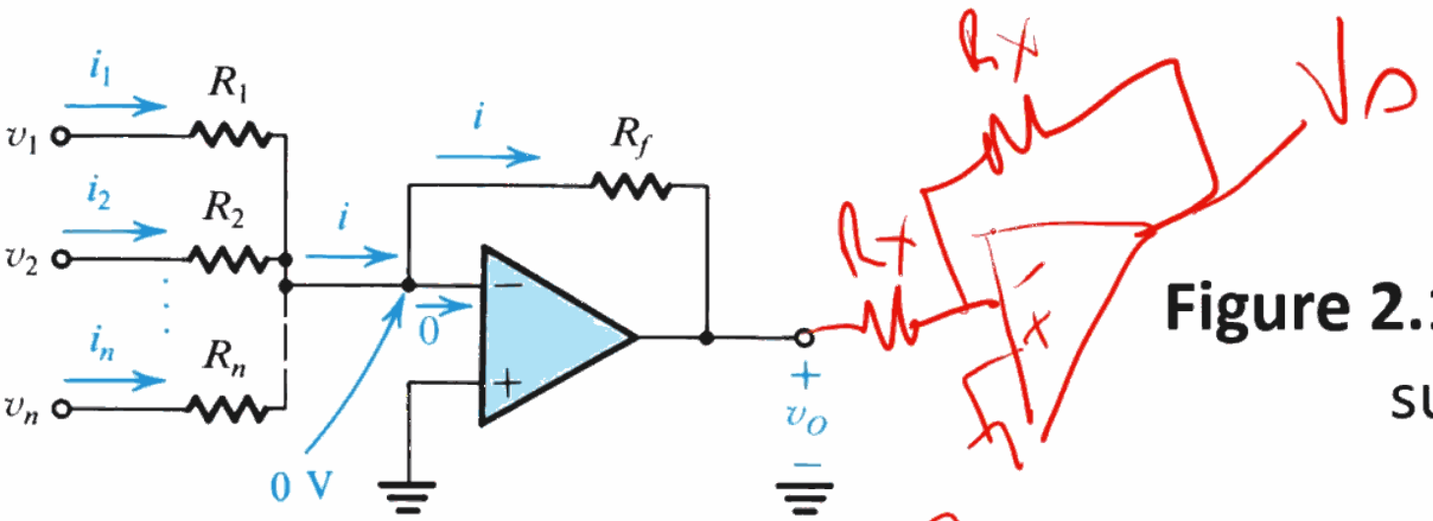


Figure 2.14: (**a**) The unity-gain buffer or follower amplifier. (**b**) Its equivalent circuit model.

$$v_{Out} = -[\,(R_f/R_{In1})v_{In1} + (R_f/R_{In2})v_{In2} + (R_f/R_{In3})v_{In3} + \ldots\,]$$



$$v_O = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \cdots + \frac{R_f}{R_n}v_n\right)$$

Figure 2.10: A weighted summer.

Figure 2.12: The non-inverting configuration.

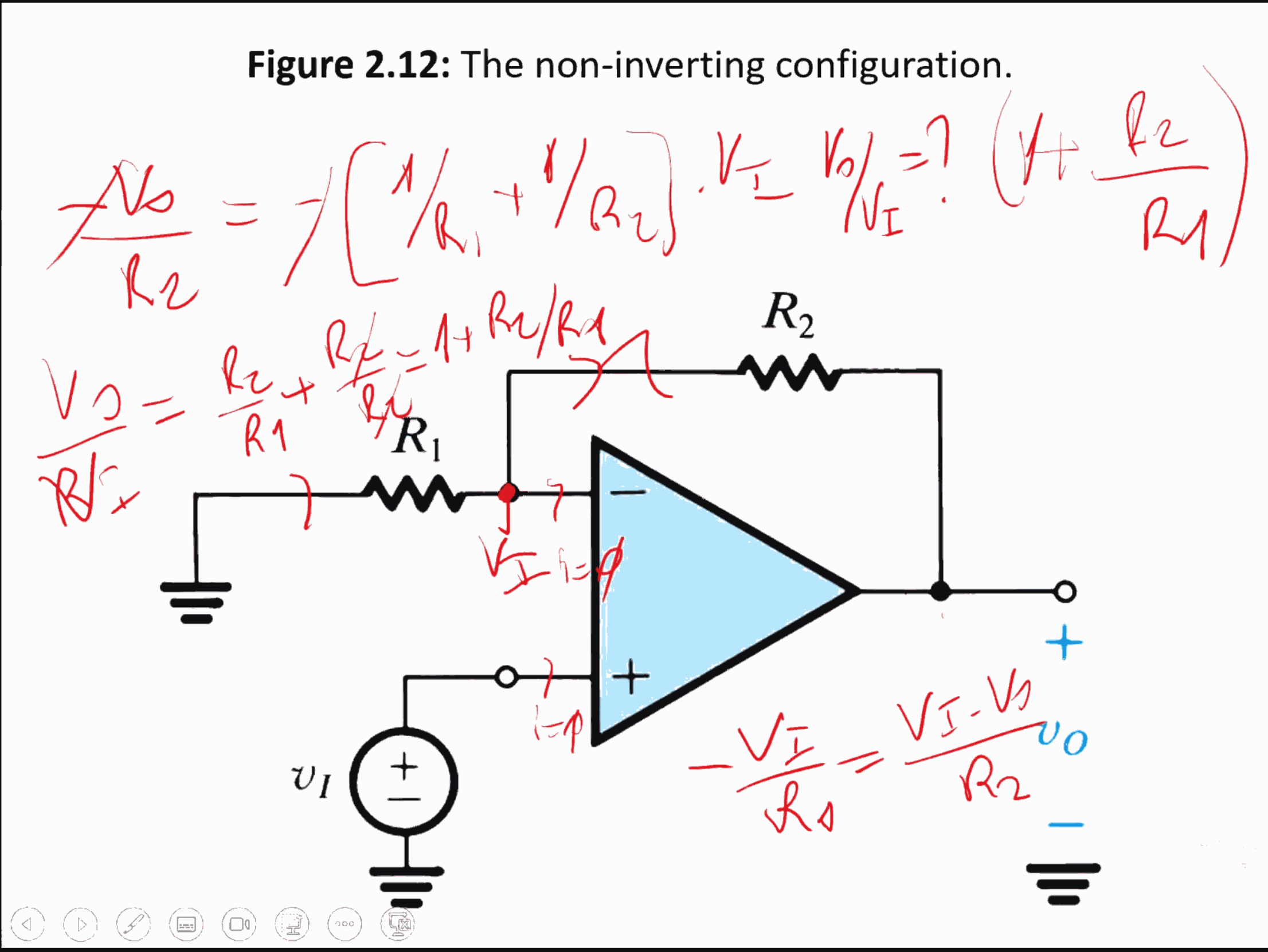


Figure 2.12: The non-inverting configuration.

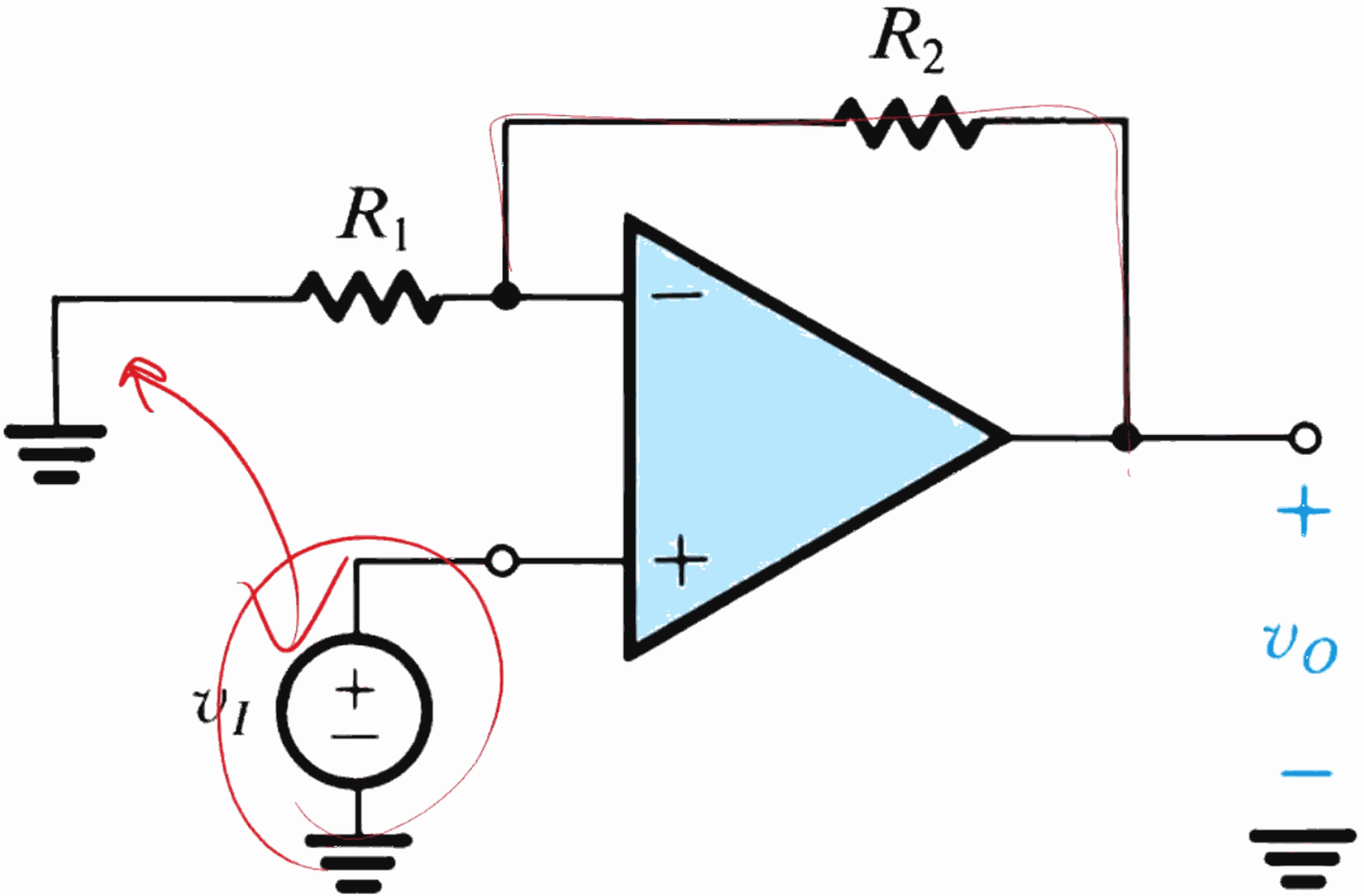
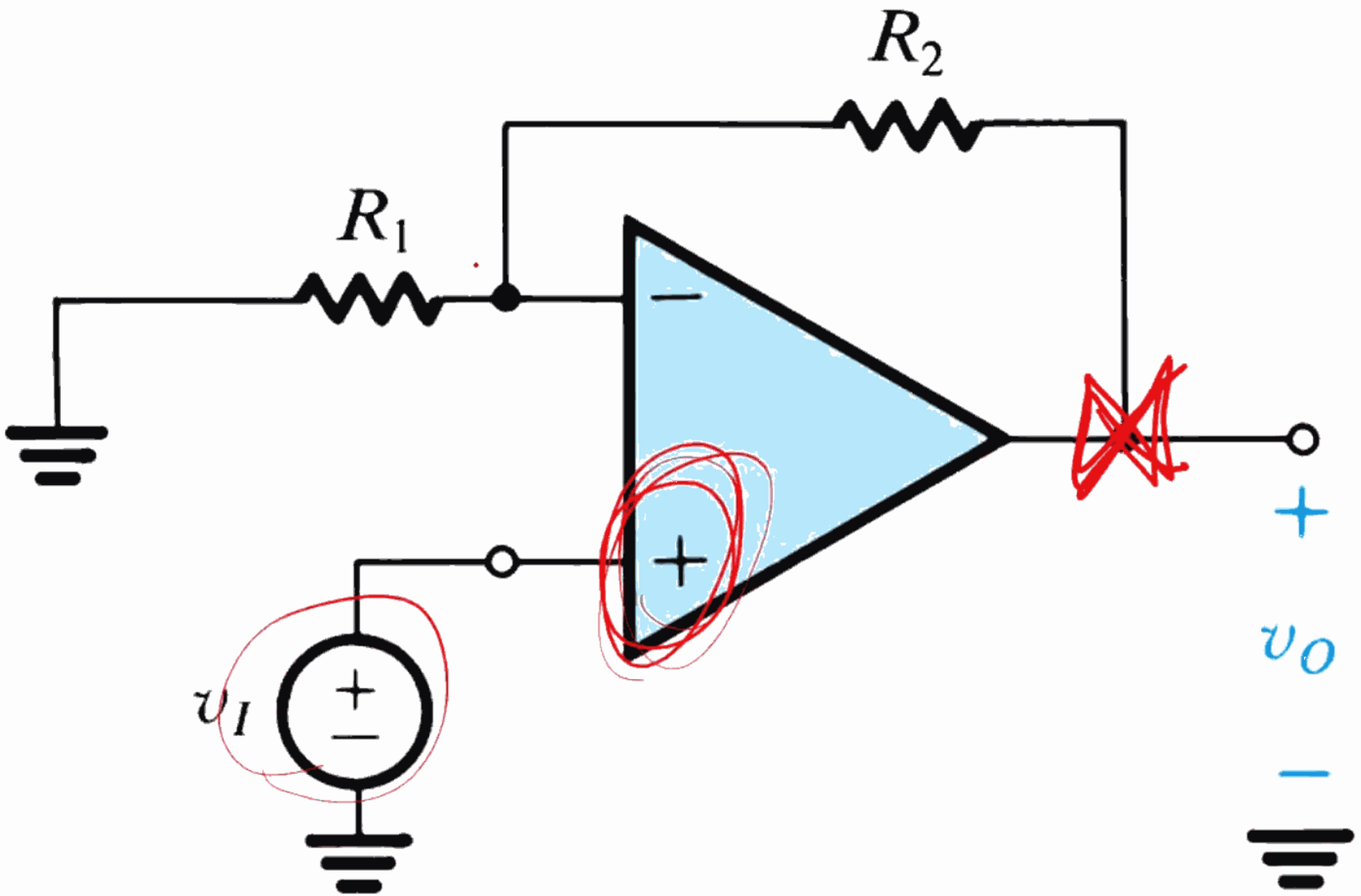


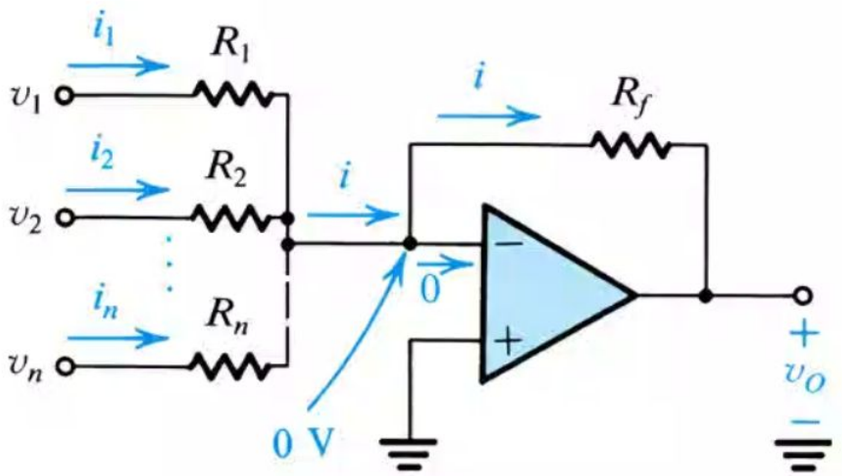
Figure 2.12: The non-inverting configuration.



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2.2.4. An Important Application – The Weighted Summer

- **weighted summer** - is a closed-loop amplifier configuration which provides an output voltage which is weighted sum of the inputs.

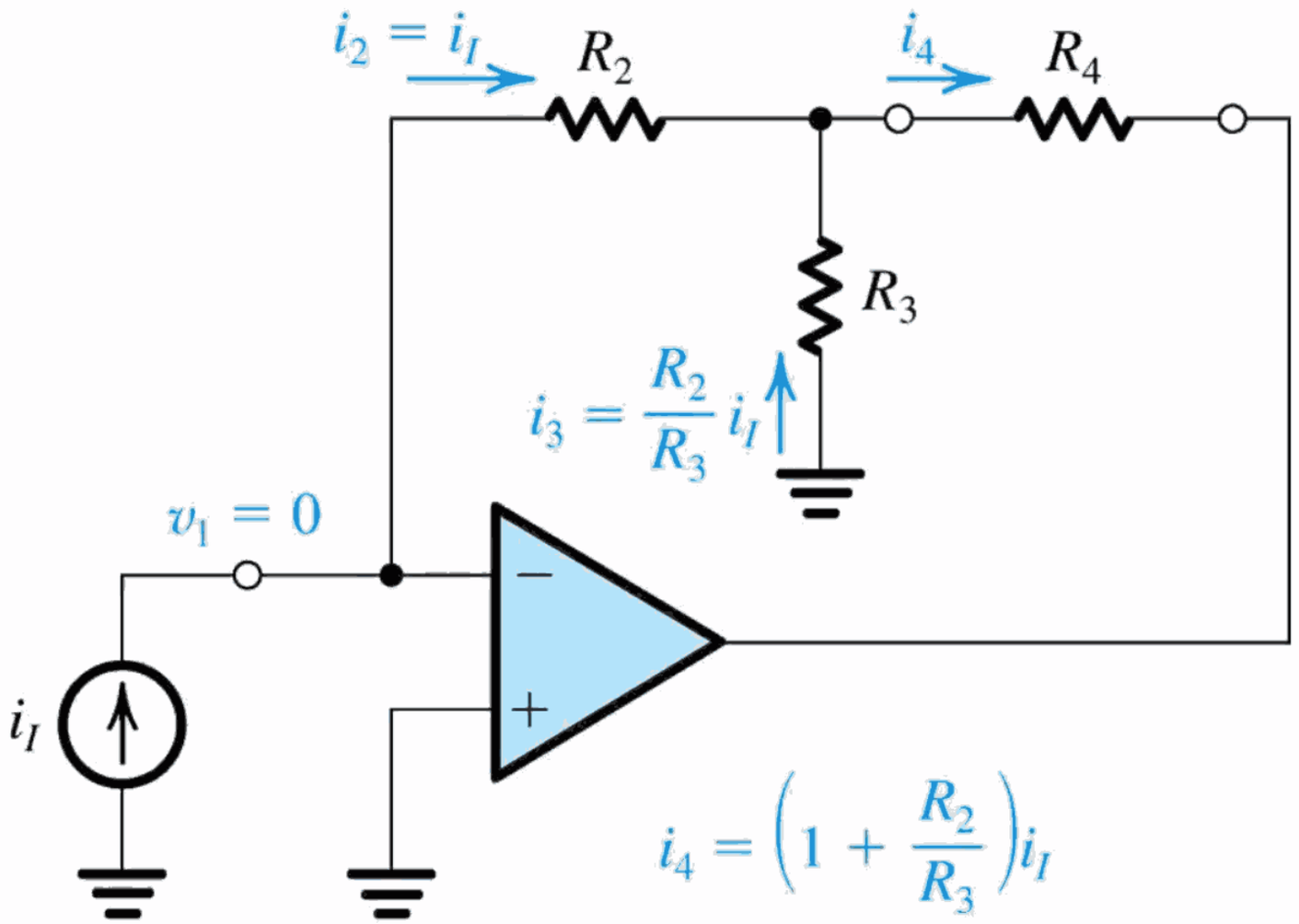


$$v_O = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \cdots + \frac{R_f}{R_n}v_n\right)$$

Figure 2.10: A weighted summer.

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Figure 2.9: A current amplifier based on the circuit of Fig. 2.8. The amplifier delivers its output current to R_4. It has a current gain of $(1 + R_2/R_3)$, a zero input resistance, and an infinite output resistance. The load (R_4), however, must be floating (i.e., neither of its two terminals can be connected to ground).



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2.2.3. Input and Output Resistances

- **Q:** What is input resistance for inverting op-amp? How is it defined mathematically?
  - **A:** R_1 (refer to math below)
- **Q:** What does this say?
  - **A:** That, for the combination of ideal op-amp and external resistors, input resistance will be finite…

this assumes that ideal op-amp and external resistors are considered "one unit"

$$R_i = \frac{v_{In}}{\underbrace{i_{In}}_{\text{same as } i_1}} = \frac{v_{In}}{(v_{In} - \underbrace{v_1}_{\text{virtual ground } =0})/R_1} = \frac{v_{In}}{v_{In}/R_1} = R_1$$

action: simplify

action: simplify

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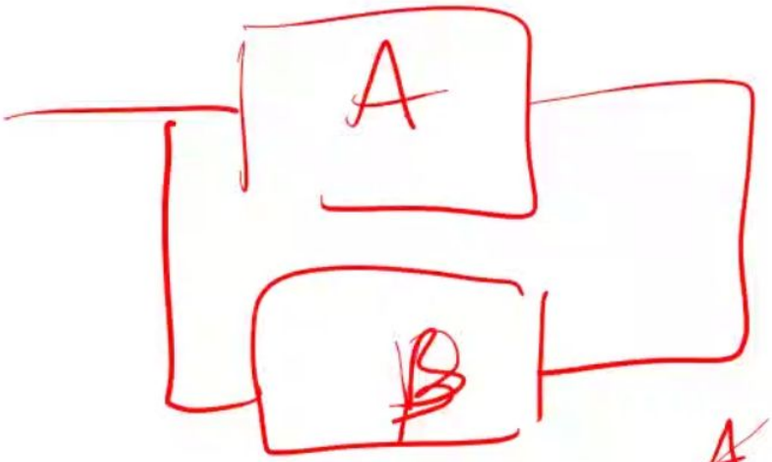
Example 2.1: Simple Inverting Amplifier

- **Problem Statement:** Consider an inverting configuration with $R_1 = 1kOhm$ and $R_2 = 100kOhm$.
- **Q(a):** Find the closed-loop gain (G) for the cases below. In each case, determine the percentage error in the magnitude of G relative to the ideal value.
  - cases are $A = 10^3, 10^4, 10^5$...
- **Q(b):** What is the voltage v_1 that appears at the inverting input terminal when $v_{In} = 0.1V$.
- **Q(c):** If the open loop gain (A) changes from 100k to 50k, what is percentage change in gain (G)?

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2.2.1. Effect of Finite Open-Loop Gain



- **Q:** Under what condition can $G = -R_2 / R_1$ be employed over the more complex expression?
  - **A:** If $1 + (R_2/R_1) << A$, then simpler expression may be used.

$$\text{if } 1+\frac{R_2}{R_1} << A \text{ then } G_{A=\infty} = -\frac{R_2}{R_1} \text{ else } G_{A<\infty} = \frac{-R_2/R_1}{1+\left(\frac{1+(R_2/R_1)}{A}\right)}$$

ideal gain

non-ideal gain

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2.2.1. Effect of Finite Open-Loop Gain

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ideal gain

non-ideal gain

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2.2.1. Effect of Finite Open-Loop Gain

- **Q:** How does the gain expression change if open loop gain (A) is not assumed to be infinite?
  - **A:** One must employ analysis similar to the previous, result is presented below…

$$G_{A<\infty} = \frac{v_{Out}}{v_{In}} = \underbrace{\frac{-R_2/R_1}{1+\left(\frac{1+(R_2/R_1)}{A}\right)}}_{\text{if } A=\infty \text{ then the previous gain expression is yielded}} \neq \overbrace{-\frac{R_2}{R_1}}^{G_{A=\infty}}$$

non-ideal gain

ideal gain

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2.2.1. Closed-Loop Gain

- **step #5:** Define v_{Out} in terms of current flowing across R_2.
- **step #6:** Substitute v_{in} / R_1 for i_1.

virtual ground

$$v_{Out} = \underbrace{(v_1)}_{\text{virtual ground}} - (i_1 R_2) = -i_1 R_2$$

$$\boxed{v_{Out} = -\frac{R_2}{R_1} v_{In}}$$

solution

note: this expression is one of the fundamentals of electronics

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More options

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2.2.1. Closed-Loop Gain

- **Q:** How does one analyze closed-loop gain for inverting configuration of an ideal op-amp?
  - **step #1:** Begin at the output terminal
  - **step #2:** If v_{Out} is finite, then differential input must equal 0
    - virtual short circuit btw v_1 and v_2
    - virtual ground exists at v_1

$$v_2 - v_1 = \frac{v_{Out}}{\underbrace{A}_{\infty}} = 0 \quad \overbrace{\text{because } A \text{ in infinite}}$$

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Figure 2.5: The inverting closed-loop configuration.

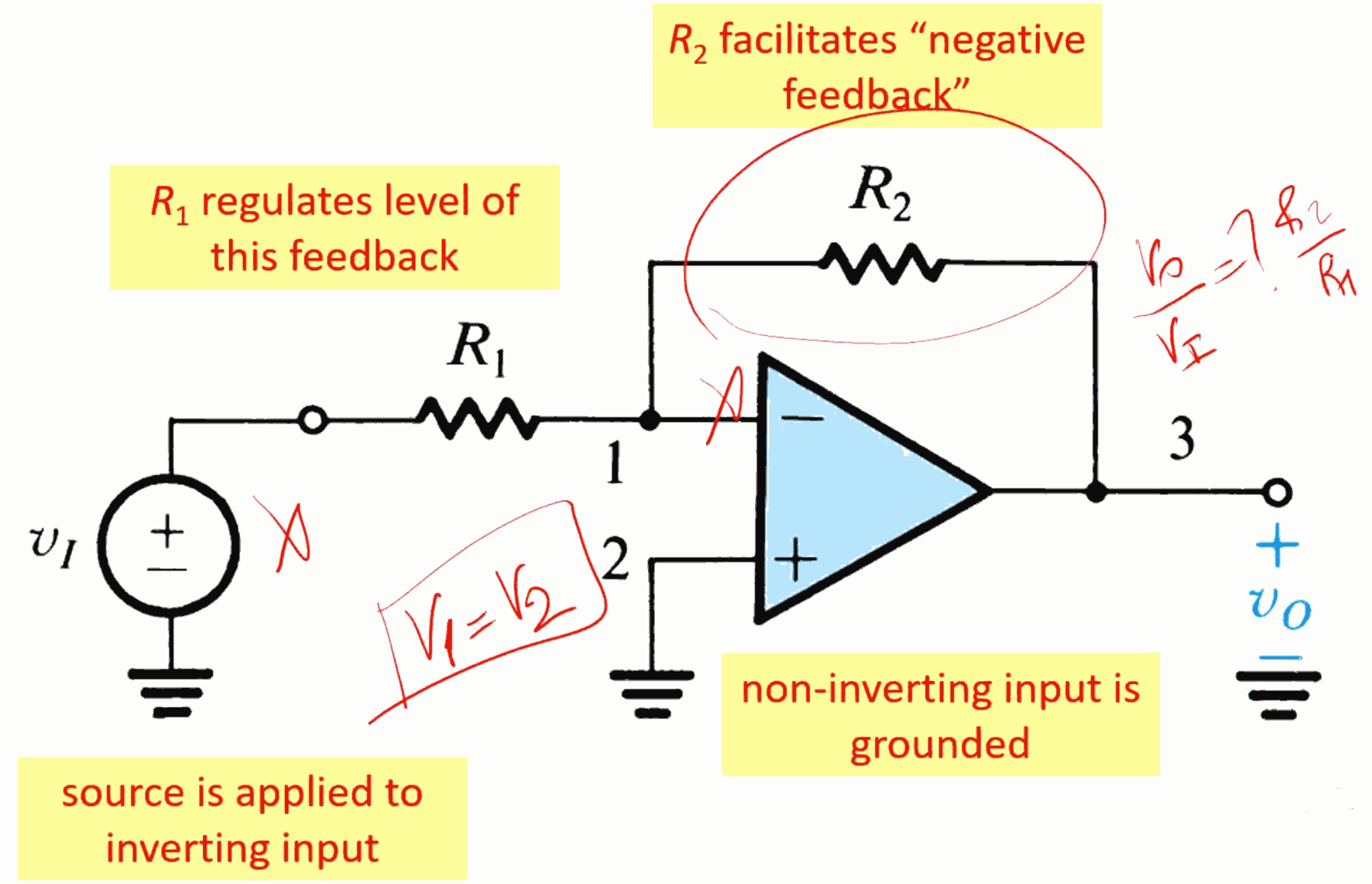
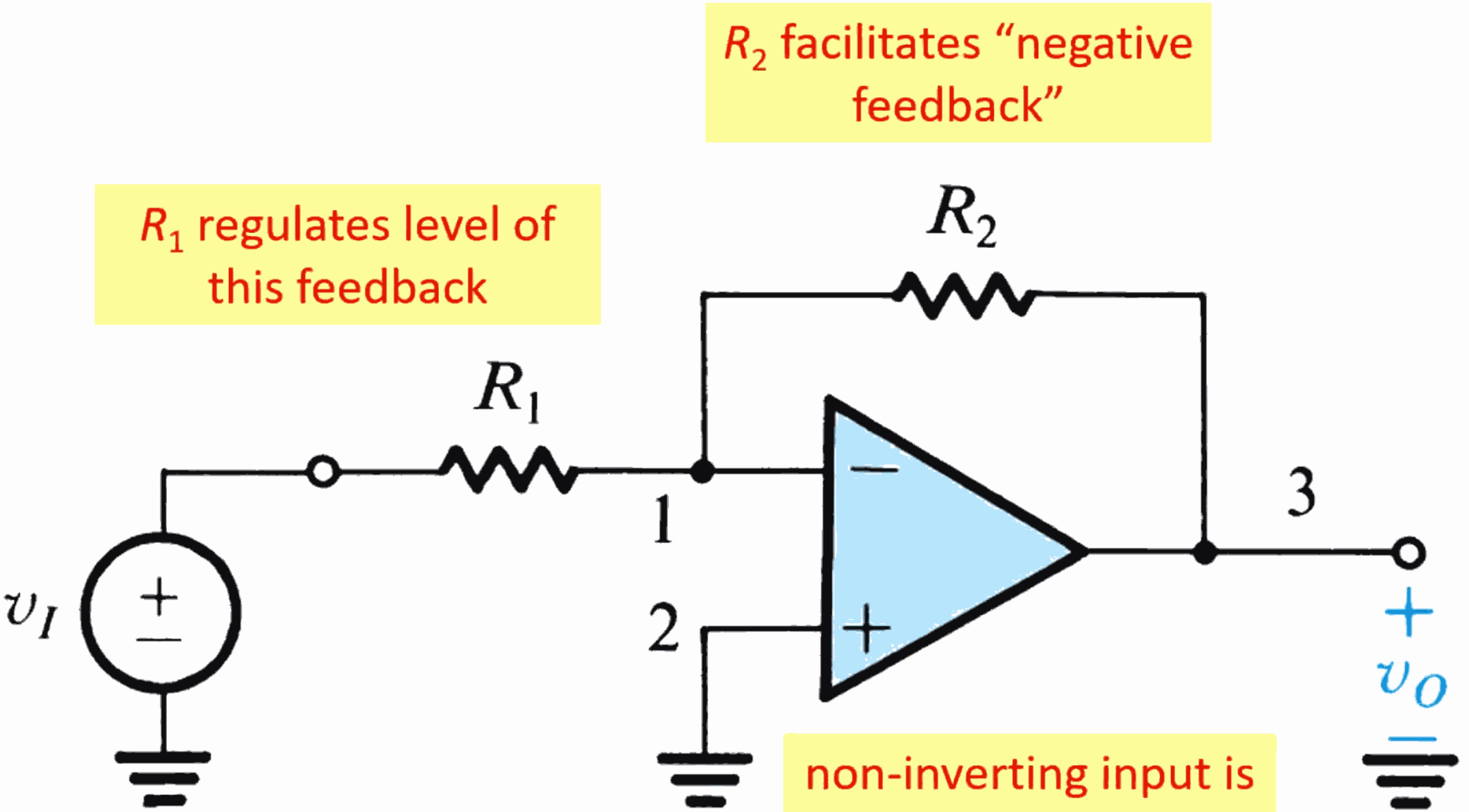


Figure 2.5: The inverting closed-loop configuration.



common-mode input

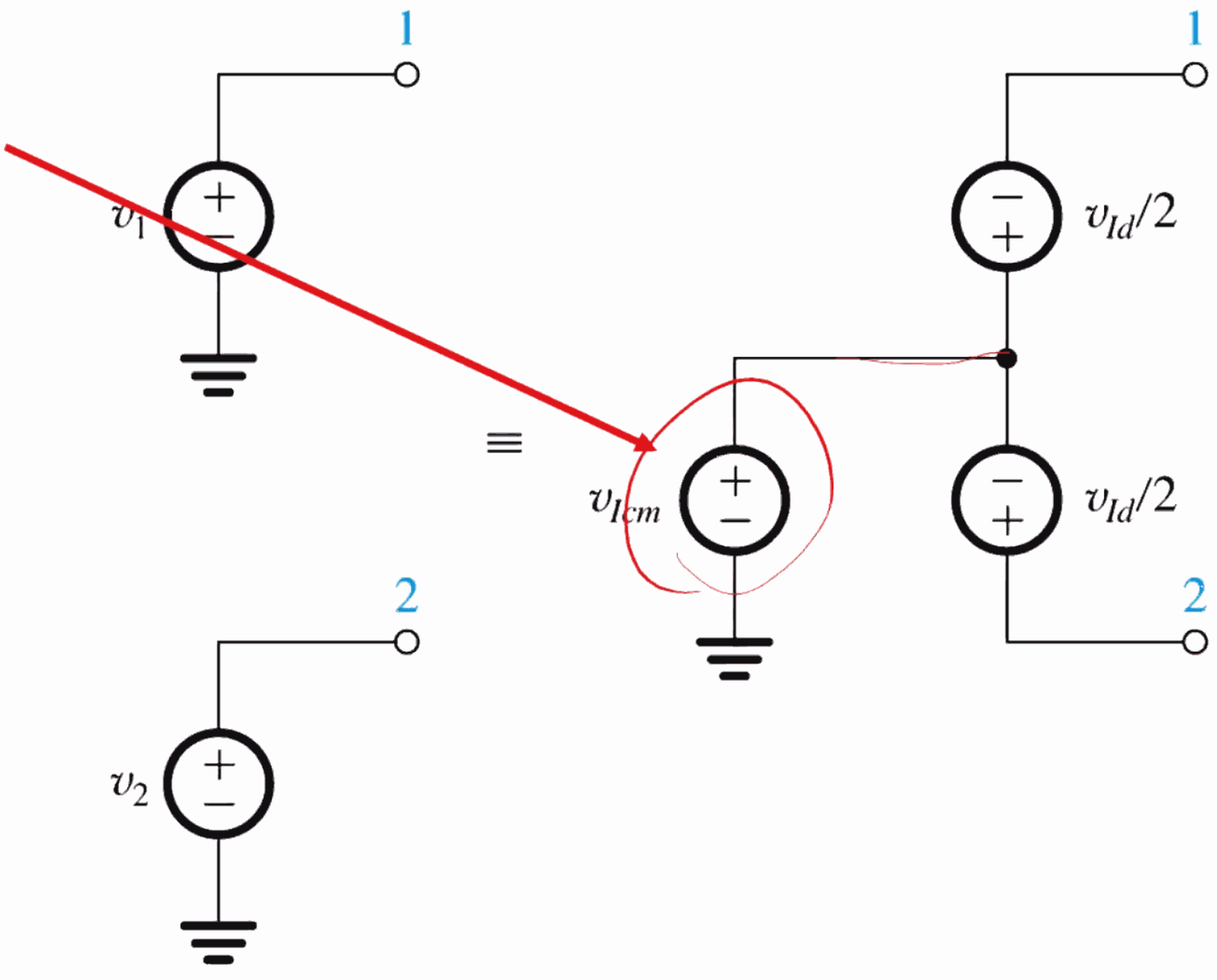
$$v_{cmi} = \frac{1}{2}(v_1 + v_2)$$

but also...

inverting input

$$v_1 = v_{cmi} - \underbrace{v_{di}}_{\text{diff}}/2$$

$$\underbrace{v_2 = v_{cmi} + v_{di}/2}_{\text{non-inverting input}}$$



2.1.3. Differential & Common-Mode Signals

- **Q:** How is common-mode input (v_{cmi}) defined in terms of v_1 and v_2?

$$\overbrace{v_{cmi} = \frac{1}{2}(v_1 + v_2)}^{\text{common-mode input}}$$

but also...

$$\overbrace{v_1 = v_{cmi} - \underbrace{v_{di}}_{\text{diff}}/2}^{\text{inverting input}}$$

$$\underbrace{v_2 = v_{cmi} + v_{di}/2}_{\text{non-inverting input}}$$

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2.1.2. Function and Characteristics of Ideal Op Amp

- **Table 2.1: Characteristics of Ideal Op Amp**
  - infinite input impedance
  - zero output impedance
  - zero common-mode gain ($A_{cm} = 0$)
    - complete common-mode rejection
  - infinite open-loop gain (A = infinity)
  - infinite bandwidth

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2.1.2. Function and Characteristics of Ideal Op-Amp

- Similarly, two components of gain exist…
  - **differential gain** (A) – gain applied to differential input ONLY
  - **common-mode gain** (A_{cm}) – gain applied to common-mode input ONLY

$$v_{out} = \overbrace{\left(A_{cm}10 + A1\right)}^{\text{e.g. } v_1 = 10+1} - \overbrace{\left(A_{cm}10 - A1\right)}^{\text{e.g. } v_2 = 10-1} = \overbrace{A_{cm}\left(10-10\right)}^{\text{common-mode output}} + \overbrace{A\left(1+1\right)}^{\text{differential output}}$$

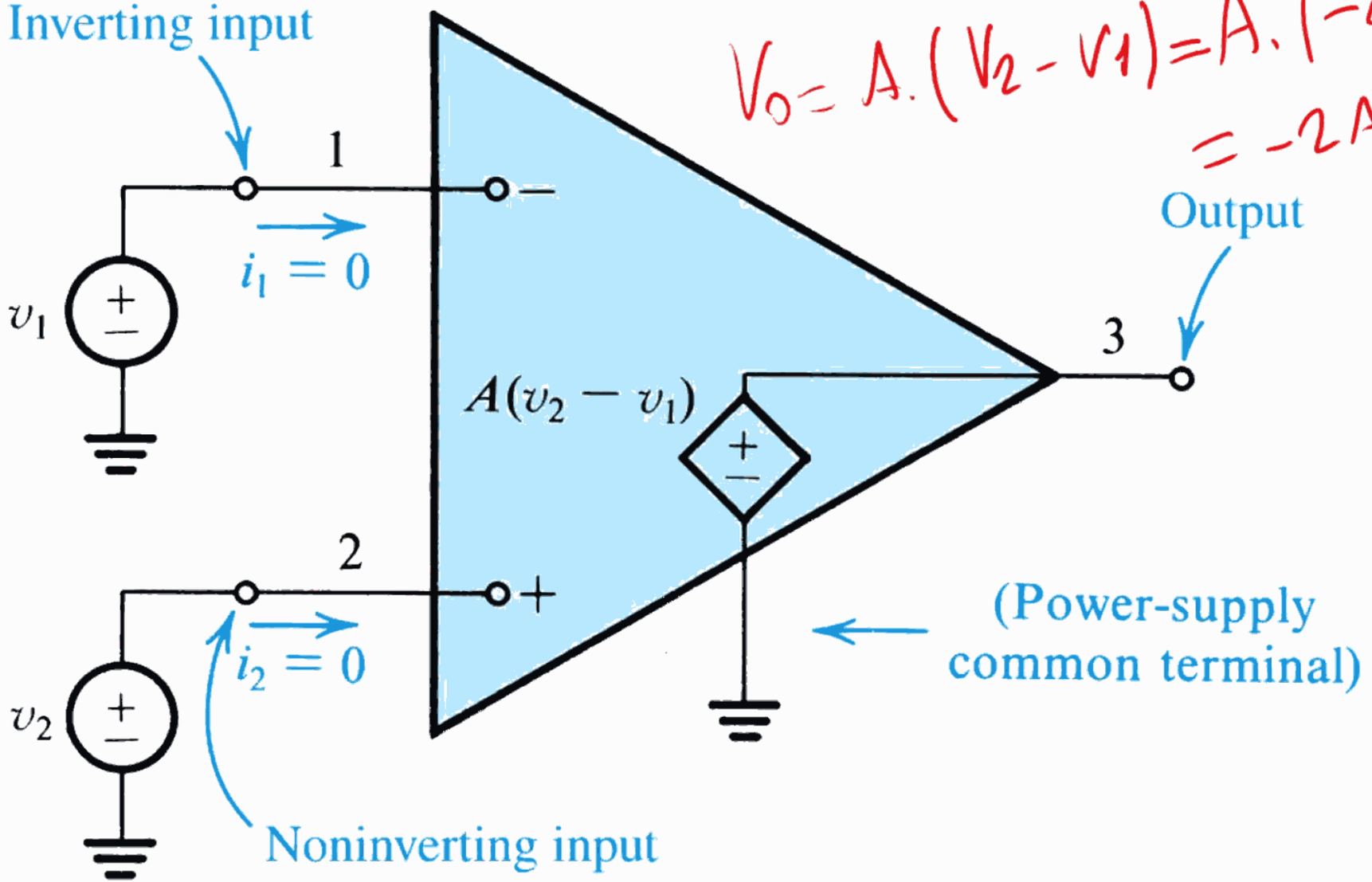
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$V_1 = 0 + \sin \omega t$

$V_2 = 0 - \sin \omega t$

$V_o = A.(V_2 - V_1) = A.(-2\sin \omega t)$

$= -2A \sin \omega t$



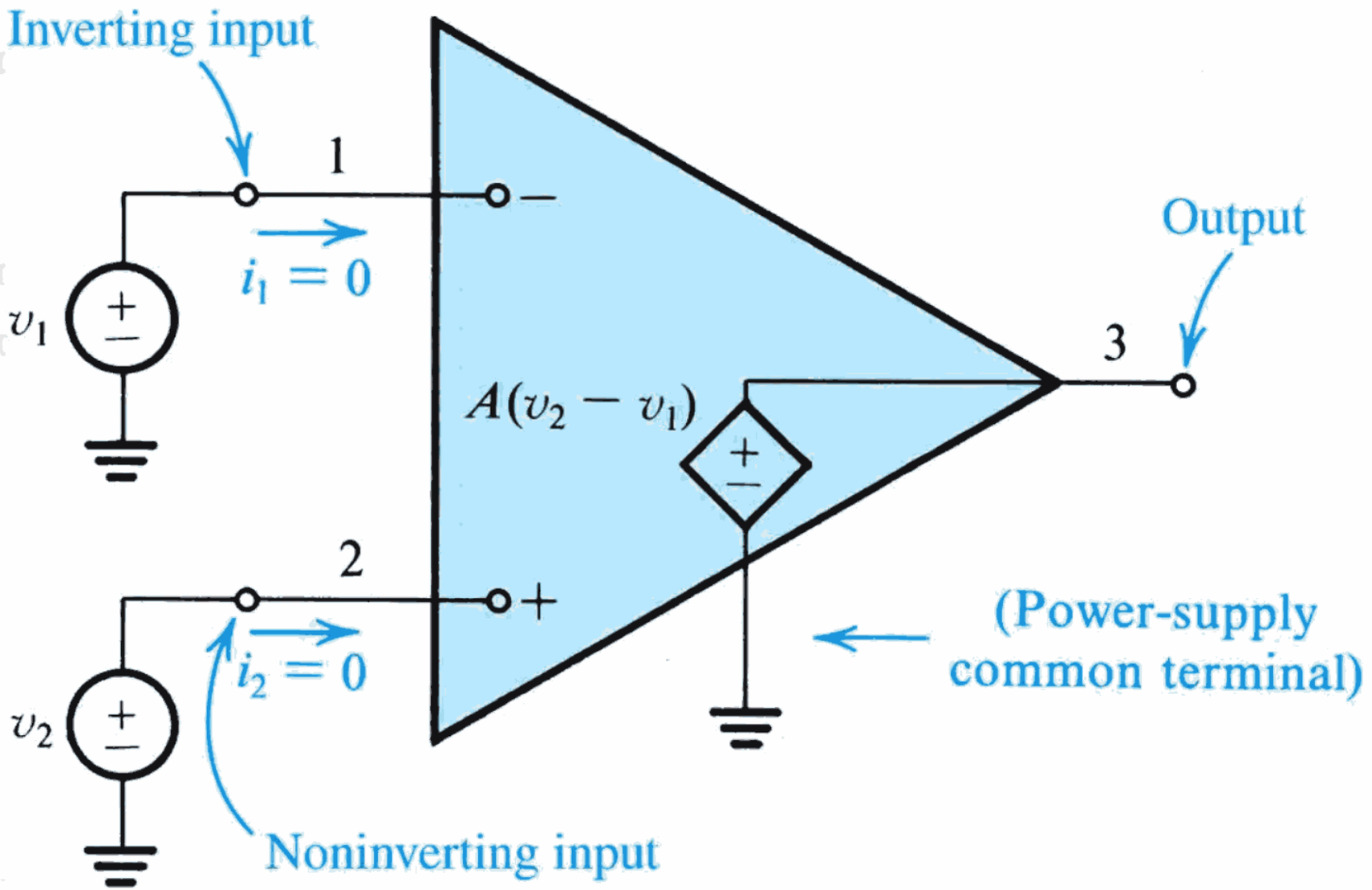
2.1.2. Function and Characteristics of Ideal Op-Amp

- An amplifier's input is composed of two components…
  - **differential input** (v_{dfi}) – is difference between inputs at inverting and non-inverting terminals
  - **common-mode input** (v_{cmi}) – is input present at both inverting and non-inverting terminals

$$v_{in} = (10+1)-(10-1) = \overbrace{(10-10)}^{\text{common-mode input } (v_{cmi})} + \overbrace{(1+1)}^{\text{differential input } (v_{dfi})}$$

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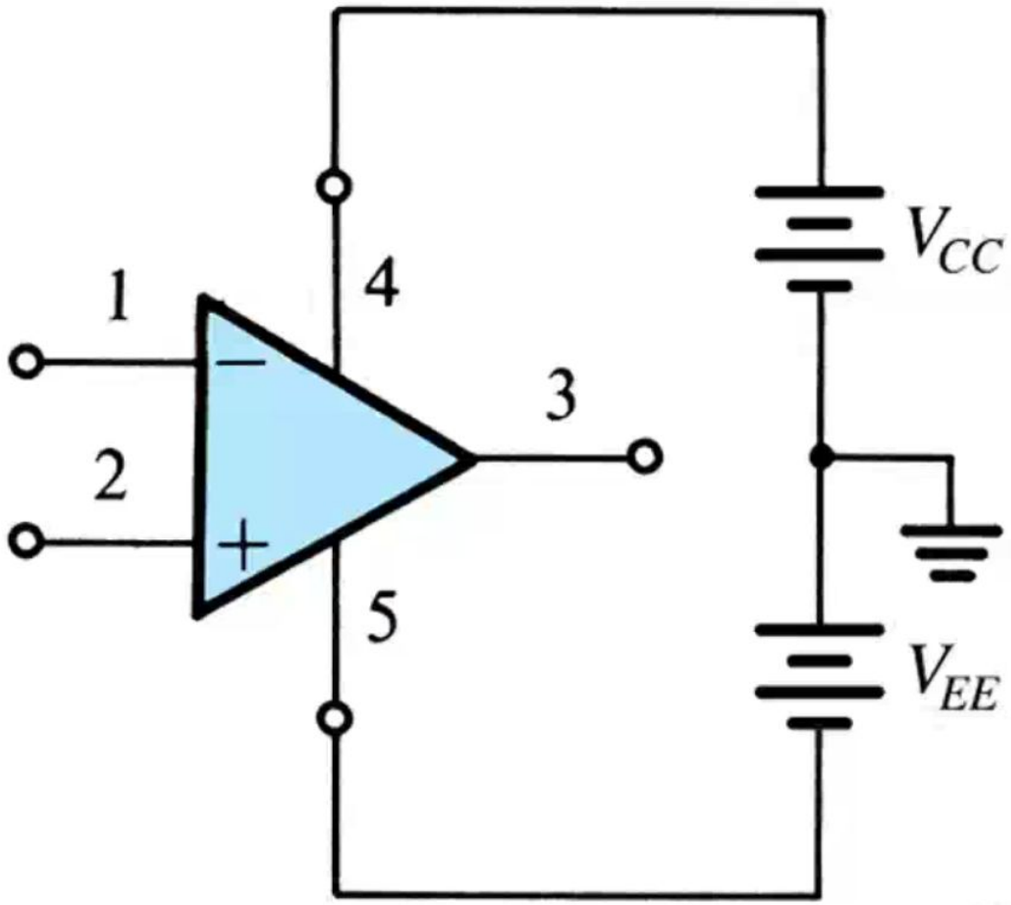
2.1.2. Function and Characteristics of Ideal Op Amp



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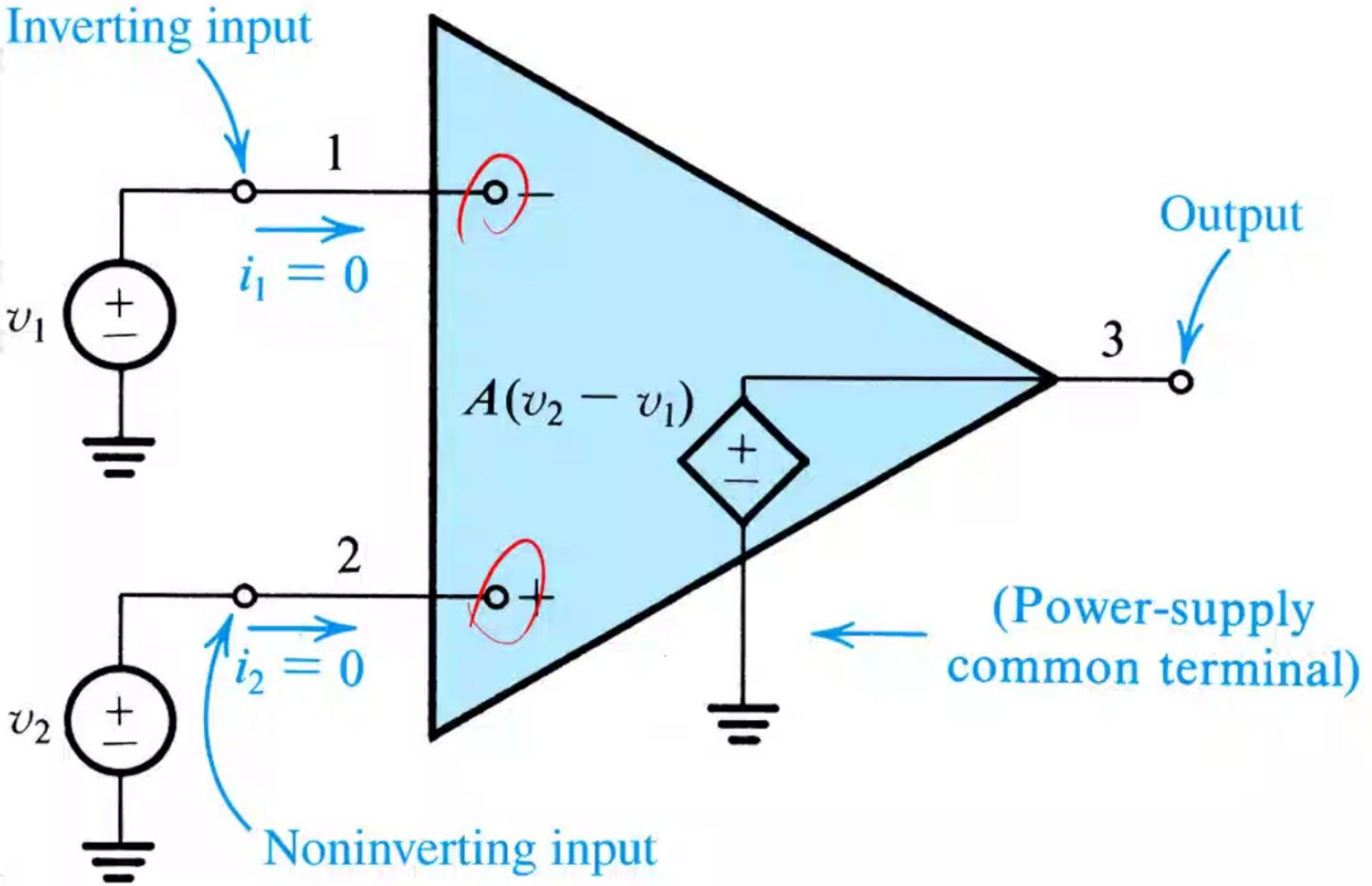
2.1.1. The Op Amp Terminals

- **terminal #1**
  - inverting input
- **terminal #2**
  - non-inverting input
- **terminal #3**
  - output
- **terminal #4**
  - positive supply V_{CC}
- **terminal #5**
  - negative supply V_{EE}



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2.1.2. Function and Characteristics of Ideal Op Amp



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2.1.2. Function and Characteristics of Ideal Op Amp

$V_3 = A(V_+ - V_-)$

- ideal gain is defined below

$$v_3 = A(v_2 - v_1)$$

- ideal input characteristic is infinite impedance
- ideal output characteristic is zero impedance
- differential gain (A) is infinite
- bandwidth gain is constant from dc to high frequencies

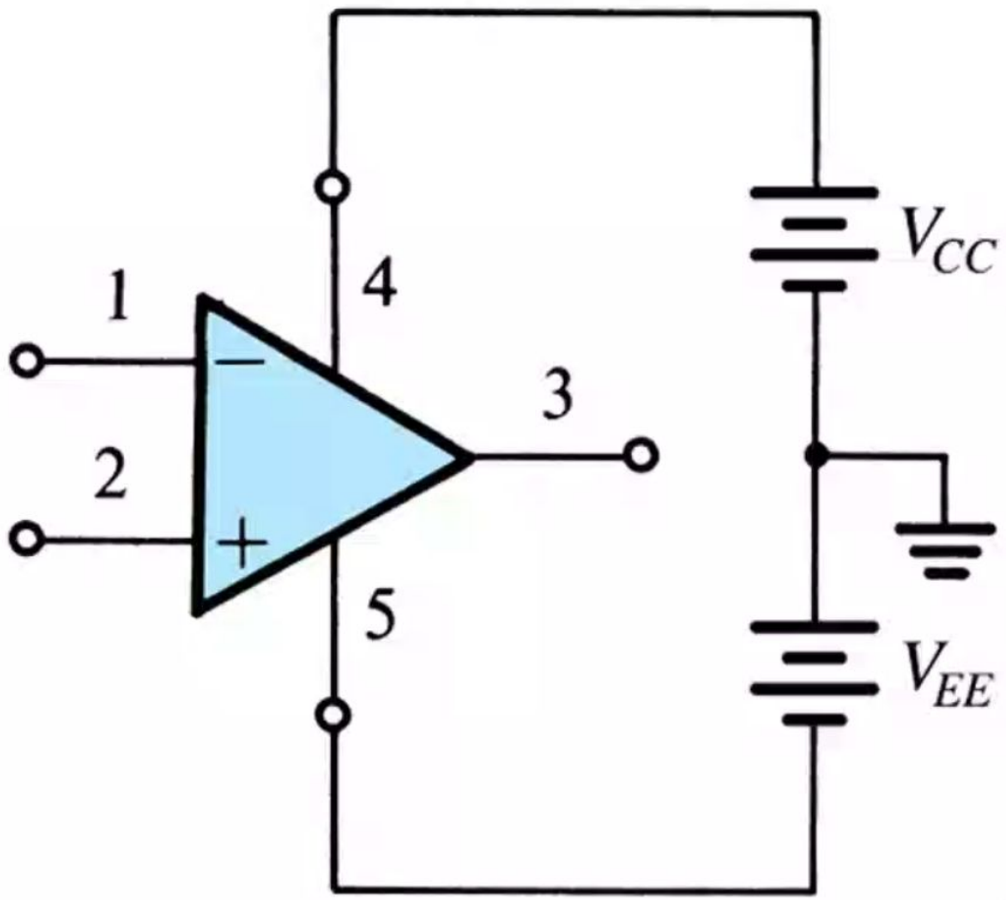
Q: But, is an amplifier with infinite gain of any use?

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2.1.1. The Op Amp Terminals

- **terminal #1**
  - inverting input
- **terminal #2**
  - non-inverting input
- **terminal #3**
  - output
- **terminal #4**
  - positive supply V_{CC}
- **terminal #5**
  - negative supply V_{EE}



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Introduction

- **IN THIS CHAPTER YOU WILL LEARN**
  - The terminal characteristics of the ideal op-amp.
  - How to analyze circuits containing op-amps, resistors, and capacitors.
  - How to use op-amps to design amplifiers having precise characteristics.

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Conclusion (6)

- Single-time-constant (STC) networks are those networks that are composed of, or may be reduced to, one reactive component (*L* or *C*) and one resistance. The time constant (τ) is *L*/*R* or *RC*.
- STC networks can be classified into two categories: low-pass (LP) and high-pass (HP). LP network pass dc and low-frequencies while attenuating high-frequencies. The opposite is true for HP.
- The gain of an LP (HP) STC circuit drops by 3*dB* below the zero-frequency (infinite-frequency) value at a frequency $\omega_0 = 1/\tau$. At high-frequencies (low-frequencies) the gain falls of at a rate of 6*dB*/octave or 20*dB*/decade.
  - Refer to Table 1.2. on page 34 and Figs. 1.23 and 1.24. Further details are provided in Appendix E.

Conclusion (5)

- The sinusoid is the only signal whose waveform is unchanged through a linear circuit. Sinusoidal signals are used to measure the frequency response of amplifiers.
- The transfer function $\mathbf{T}(s) = \mathbf{V}_o(s)/\mathbf{V}_i(s)$ of a voltage amplifier may be determined from circuit analysis. Substituting $s = j\omega$ gives $\mathbf{T}(j\omega)$ whose magnitude ($|\mathbf{T}(j\omega)|$ is the magnitude response and $\phi(\omega)$ is the phase response.
- Amplifiers are classified according to the shape of their frequency response.

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Conclusion (4)

- Amplifiers increase the signal power and thus require dc power supplies for their operation.
- The amplifier voltage gain can be expressed as a ratio A_v in V/V or in decibels, $20\log|A_v|$ in dB.
- Depending on the signal to be amplified (voltage or current) and on the desired form of output signal (voltage or current) there are four basic amplifier types: voltage, current, transconductance, and transresistance. A given amplifier may be modeled by any of these configurations, in which case their parameters are related by (1.14) through (1.16) in the text.

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Conclusion

- An electrical signal source can be represented in either Thevenin form (a voltage source v_s in series with source resistance R_s) or the Norton form (a current source i_s in parallel with resistance R_s). The Thevenin voltage v_s is the open-circuit voltage between the source terminals. The Norton current i_s is equal to the short-circuit current between the source terminals. For the two representations to be equivalent, v_s and $R_s i_s$ must be equal.
- A signal can be represented either by its waveform vs time or as the sum of sinusoids. The latter representation is known as the frequency spectrum of the signal.

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1.6.5. Classification of Amps Based on Frequency Response

- **directly coupled / dc amplifiers** – allow passage of low frequencies
- **capacitively coupled amplifiers** – allow passage of high frequencies
- **tuned amplifiers** – allow passage of a "band" of frequencies

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1.6.5. Classification of Amps Based on Frequency Response

- **internal capacitances** – cause the falloff of gain at high frequencies
  - like those seen in previous example
- **coupling capacitors** – cause the falloff of gain at low frequencies
  - are placed in between amplifier stages
  - generally chosen to be large

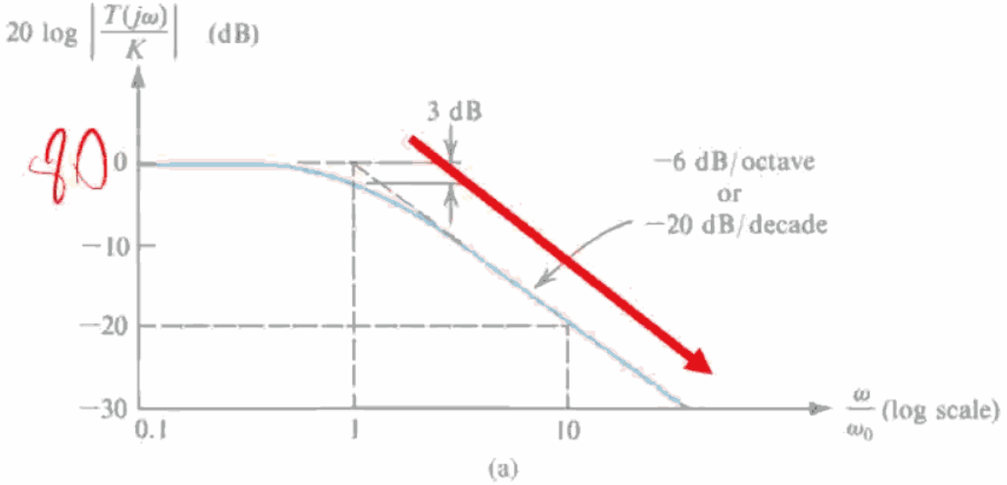
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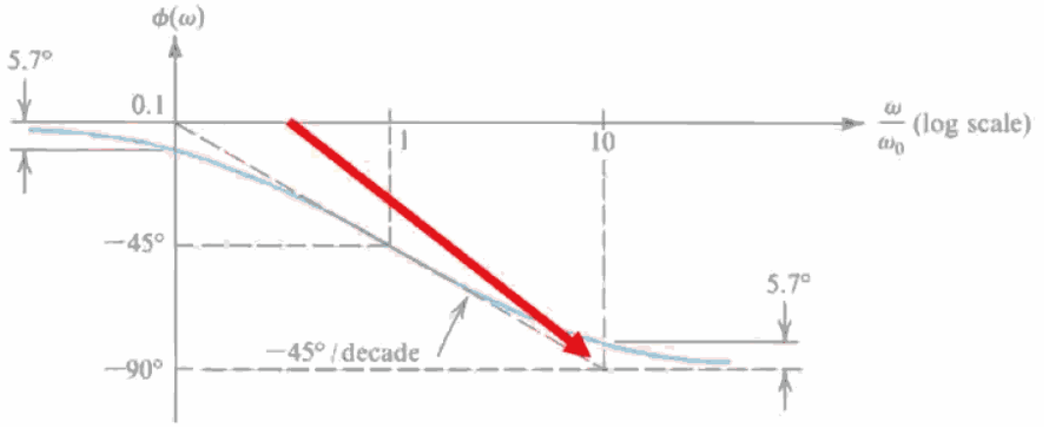
Example 1.5:
Voltage Amplifier

- Examine voltage amplifier with:
  - input resistance (R_i)
  - input capacitance (C_i)
  - gain factor (μ)
  - output resistance (R_o)
- **Q(a):** Derive an expression for the amplifier voltage gain V_o / V_s as a function of frequency. From this, find expressions for the dc gain and 3dB frequency.

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(a)



(b)

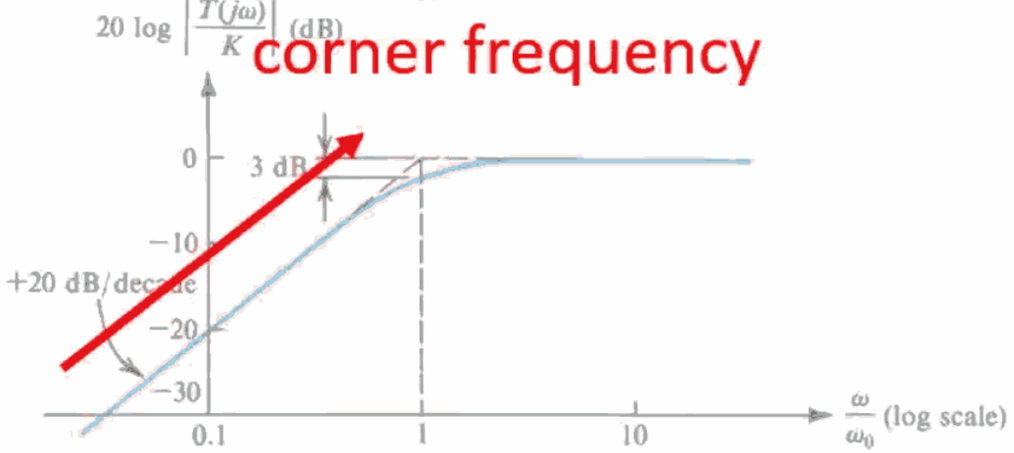
-20dB/decade drop, beginning from maximum gain at corner frequency

-45 degrees/decade drop, moving outward from -45 degree shift at corner frequency

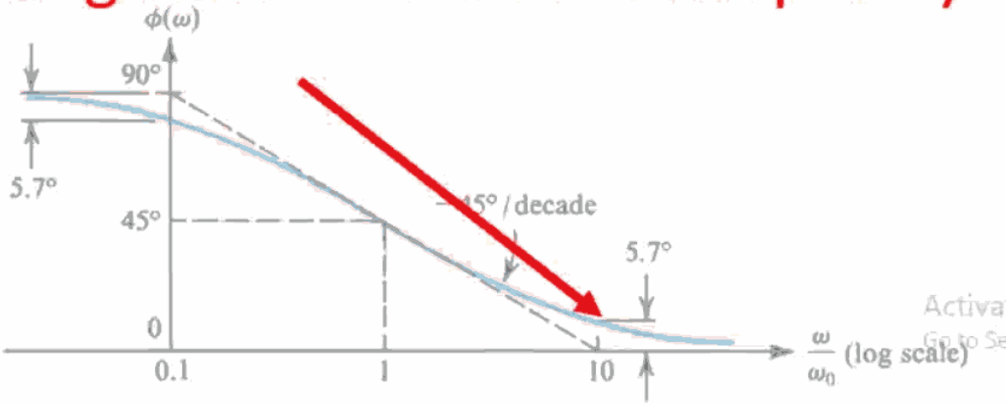
Figure: Low-Filter Magnitude (top-left) and Phase (top-right) Responses as well as High-Pass Filter (bottom-left) and Phase (bottom-right) Responses

+20dB/decade incline, until maximum gain is reached at corner frequency

-45 degrees/decade drop, moving outward from +45 degree shift at corner frequency

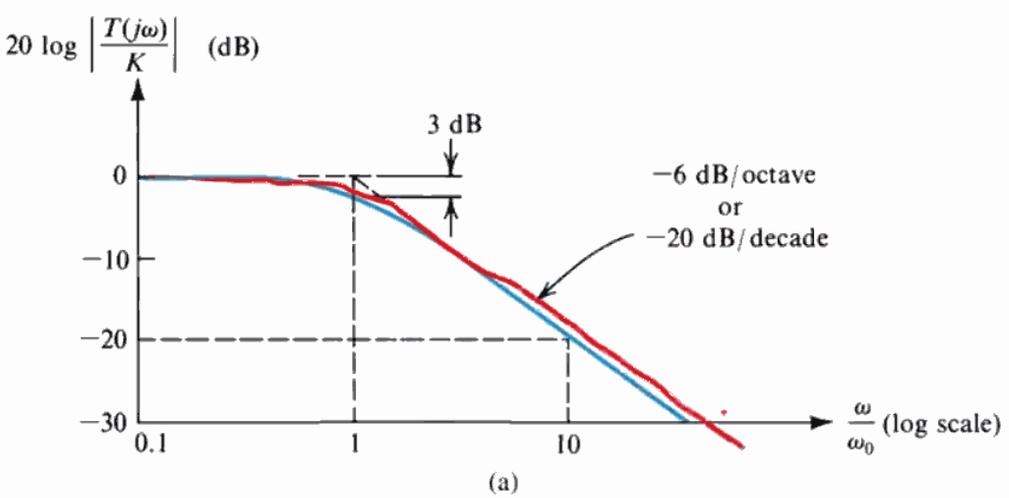


(a)



(b)

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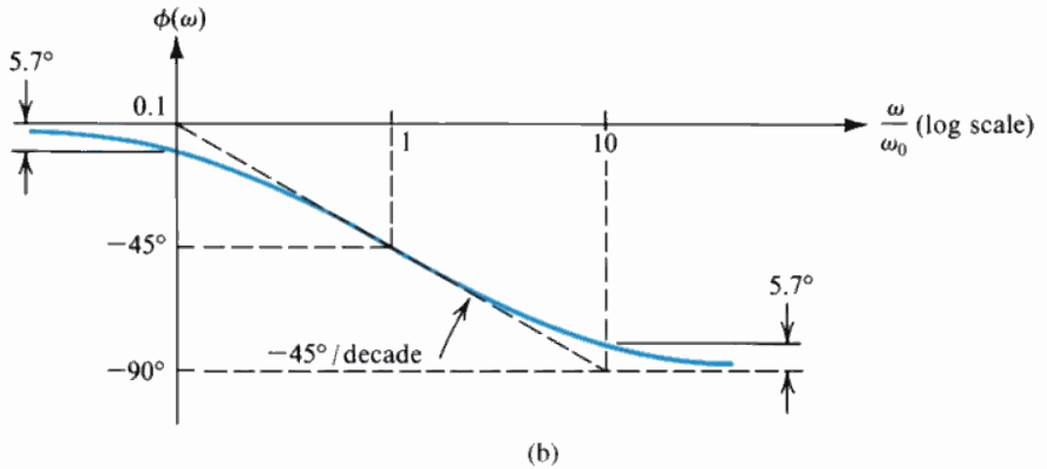
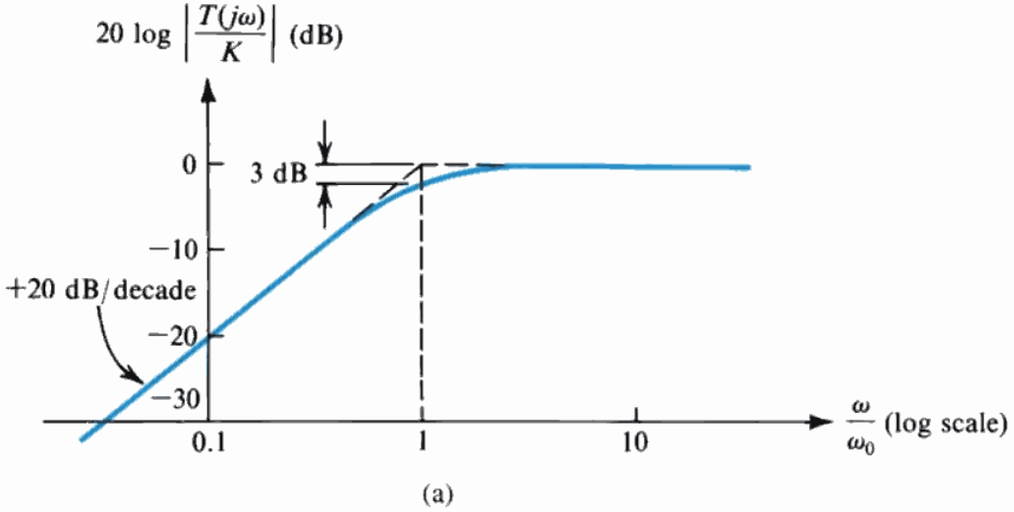
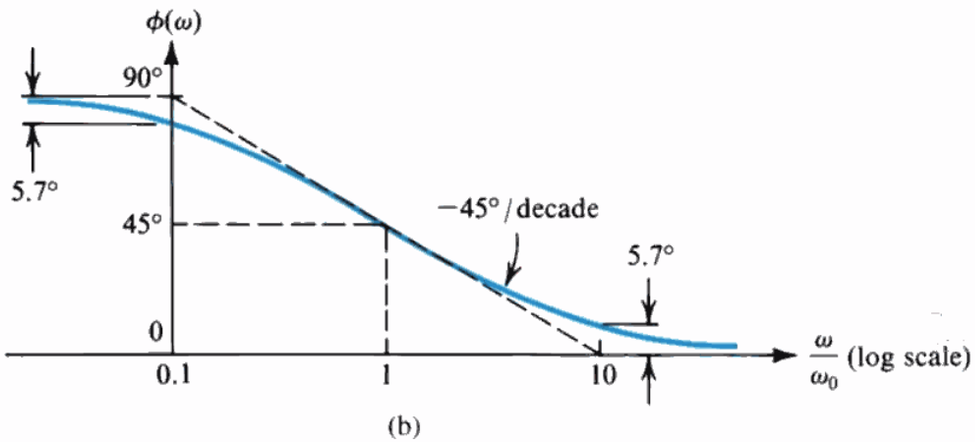


Figure: Low-Pass Filter Magnitude (top-left) and Phase (top-right) Responses as well as High-Pass Filter (bottom-left) and Phase (bottom-right) Responses





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1.6.4. Single Time-Constant Networks

Figure 1.2 : Characteristics of Various STC

	low - pass	**high - pass**
transfer function	$\frac{K}{1+(s/\omega_0)}$	$\frac{Ks}{1+\omega_0}$
transfer function (for physical freq.)	$\frac{K}{1+j(\omega/\omega_0)}$	$\frac{K}{1-j(\omega_0/\omega)}$
magnitude response	$\frac{\lvert K\rvert}{\sqrt{1+j(\omega/\omega_0)^2}}$	$\frac{\lvert K\rvert}{\sqrt{1+j(\omega_0/\omega)^2}}$
phase response	$-\tan(\omega/\omega_0)$	$\tan(\omega_0/\omega)$
transmission at $\omega=0$	K	0
transmission at $\omega=\infty$	0	K
$3db$ **Frequency**	$\omega_0=\frac{1}{\tau}$	same
Bode Plots	refer to	next slide

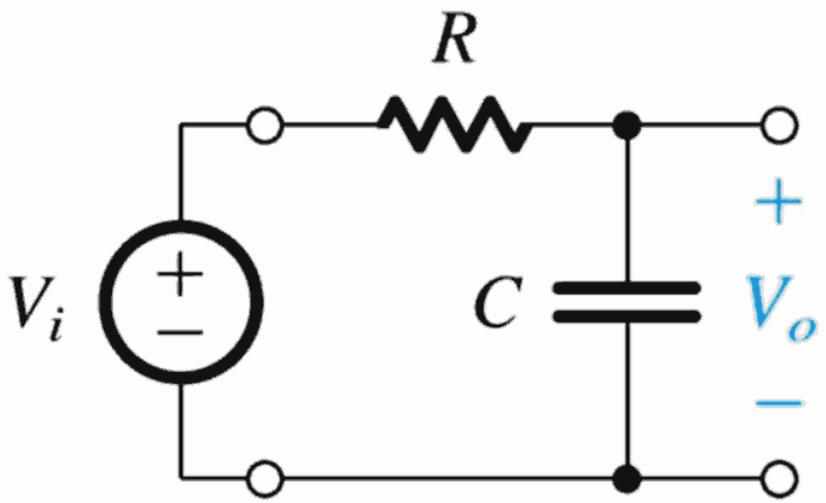
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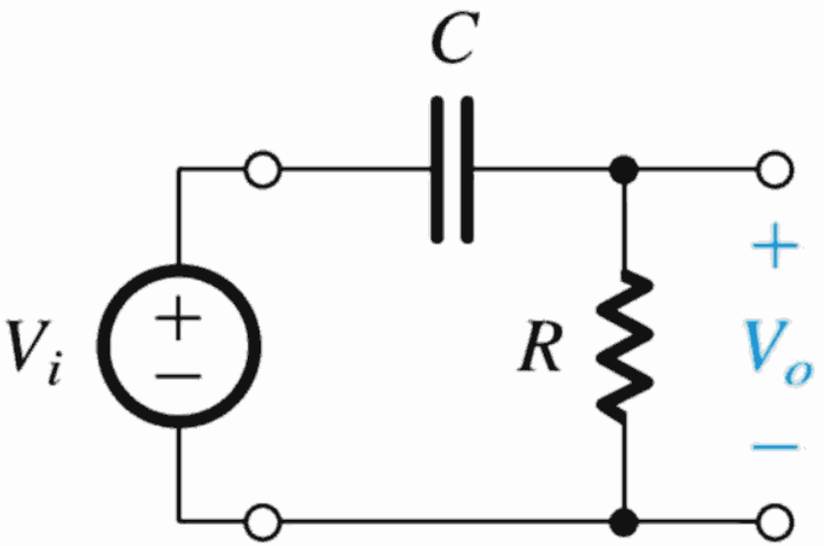
1.6.4. Single Time-Constant Networks

low-pass:
$$\frac{v_o}{v_i} = \frac{Z_o}{Z_i + Z_o} = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} \propto k\frac{1}{\omega}$$

high-pass:
$$\frac{v_o}{v_i} = \frac{Z_o}{Z_i + Z_o} = \frac{R}{R + \frac{1}{j\omega C}} \propto k\omega$$



(a)



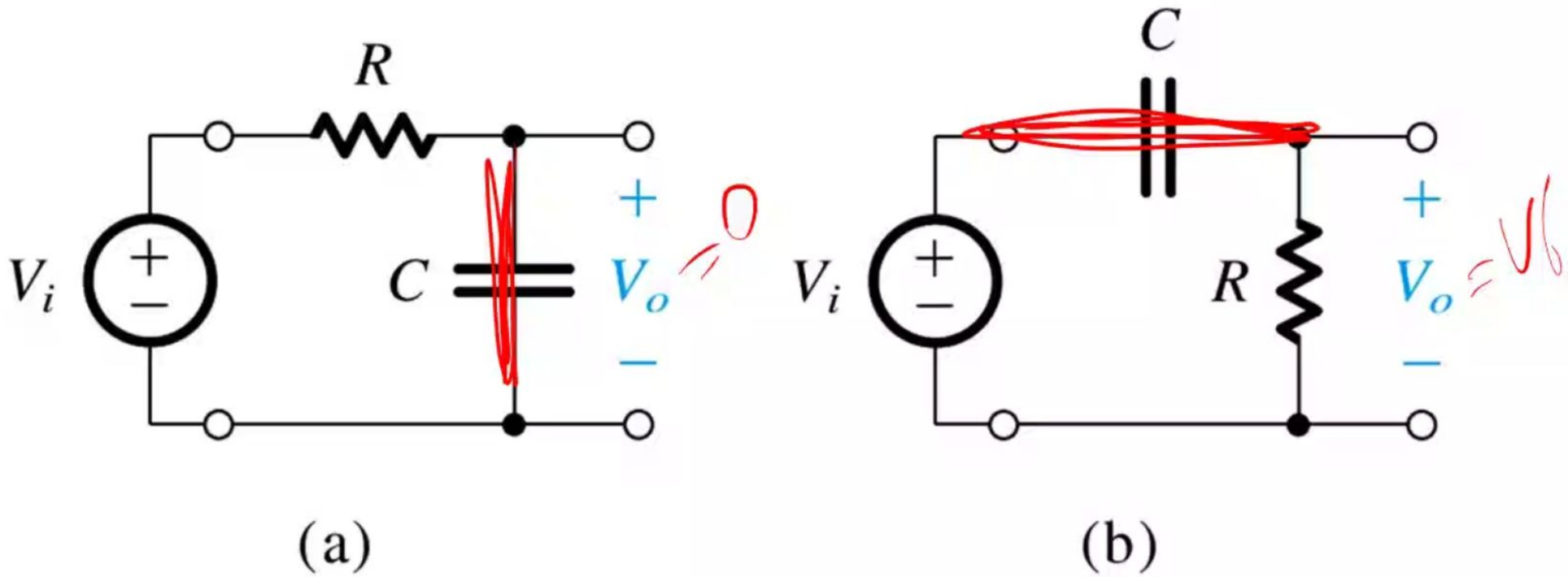
(b)

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1.6.4. Single Time-Constant Networks

- low pass filter (left)
- high pass filter (right)

Figure 1.22: Two examples of STC networks: **(a)** a low-pass network and **(b)** a high-pass network.

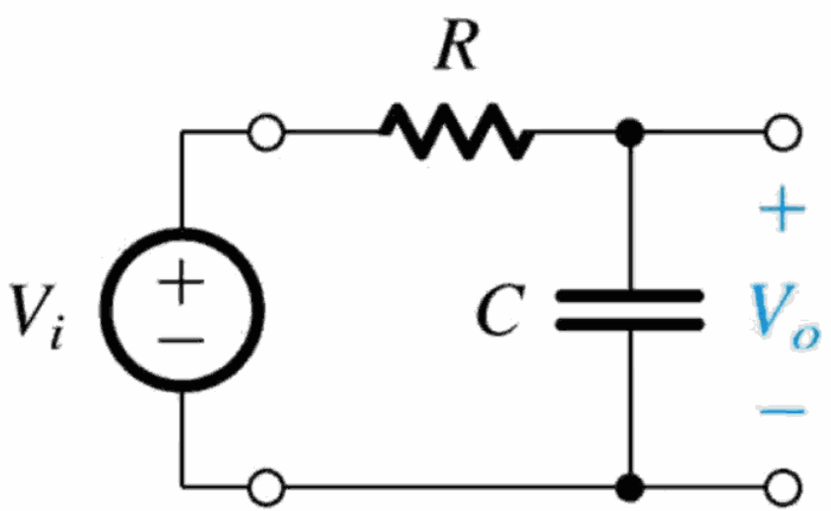


1.6.4. Single Time-Constant Networks

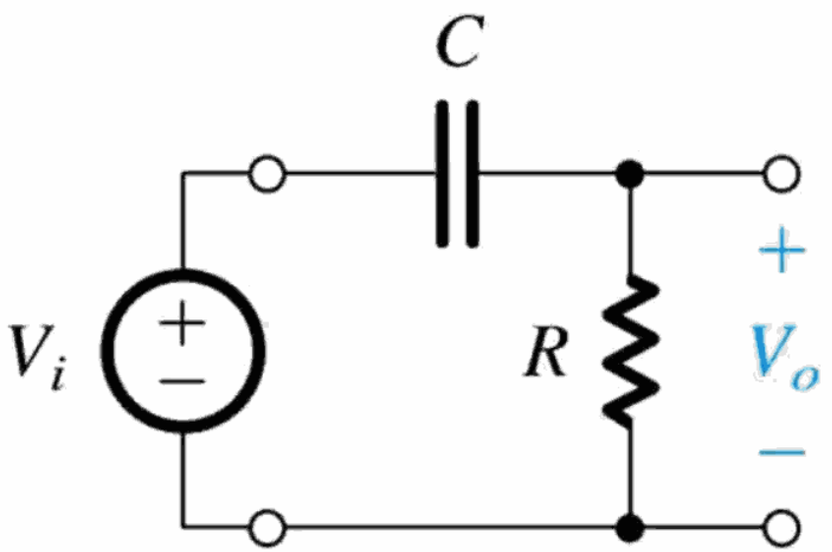
$\frac{1}{j\omega C}$

- low pass filter (left)
- high pass filter (right)

Figure 1.22: Two examples of STC networks: **(a)** a low-pass network and **(b)** a high-pass network.



(a)



(b)

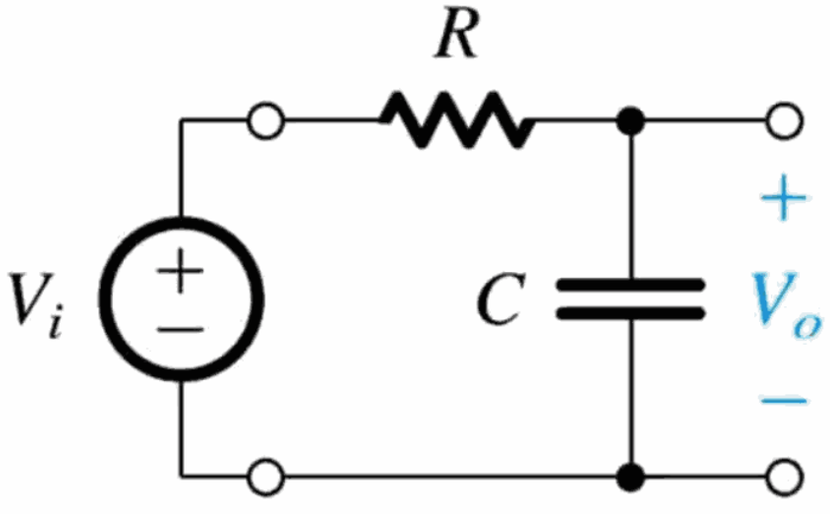
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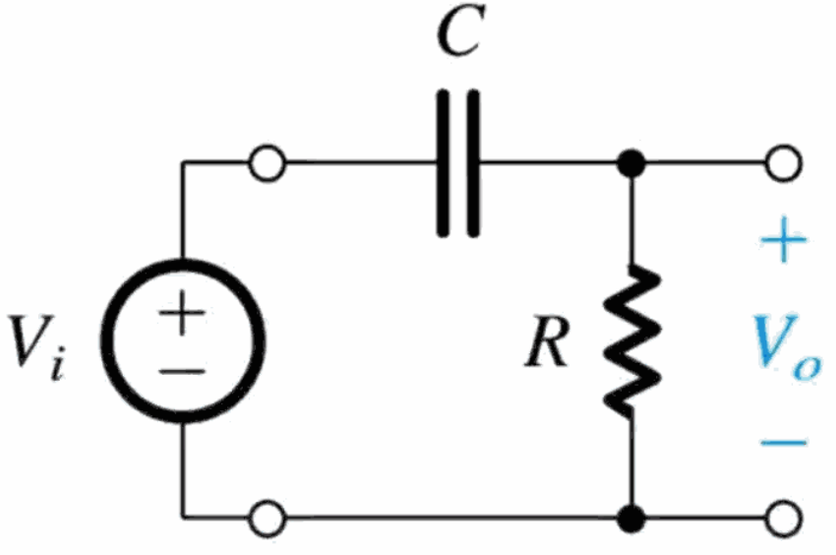
1.6.4. Single Time-Constant Networks

- low pass filter (left)
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Figure 1.22: Two examples of STC networks: **(a)** a low-pass network and **(b)** a high-pass network.

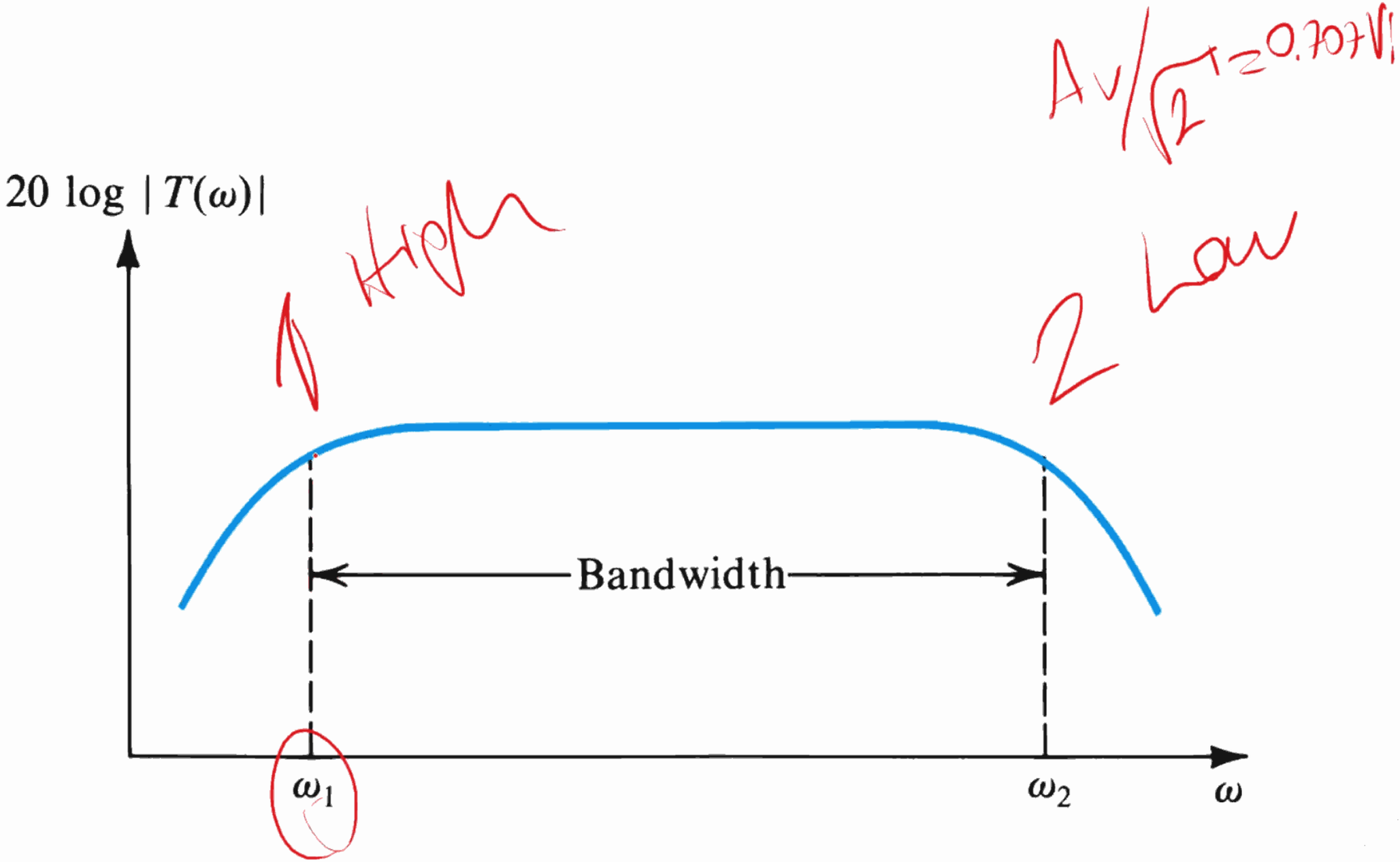


(a)



(b)

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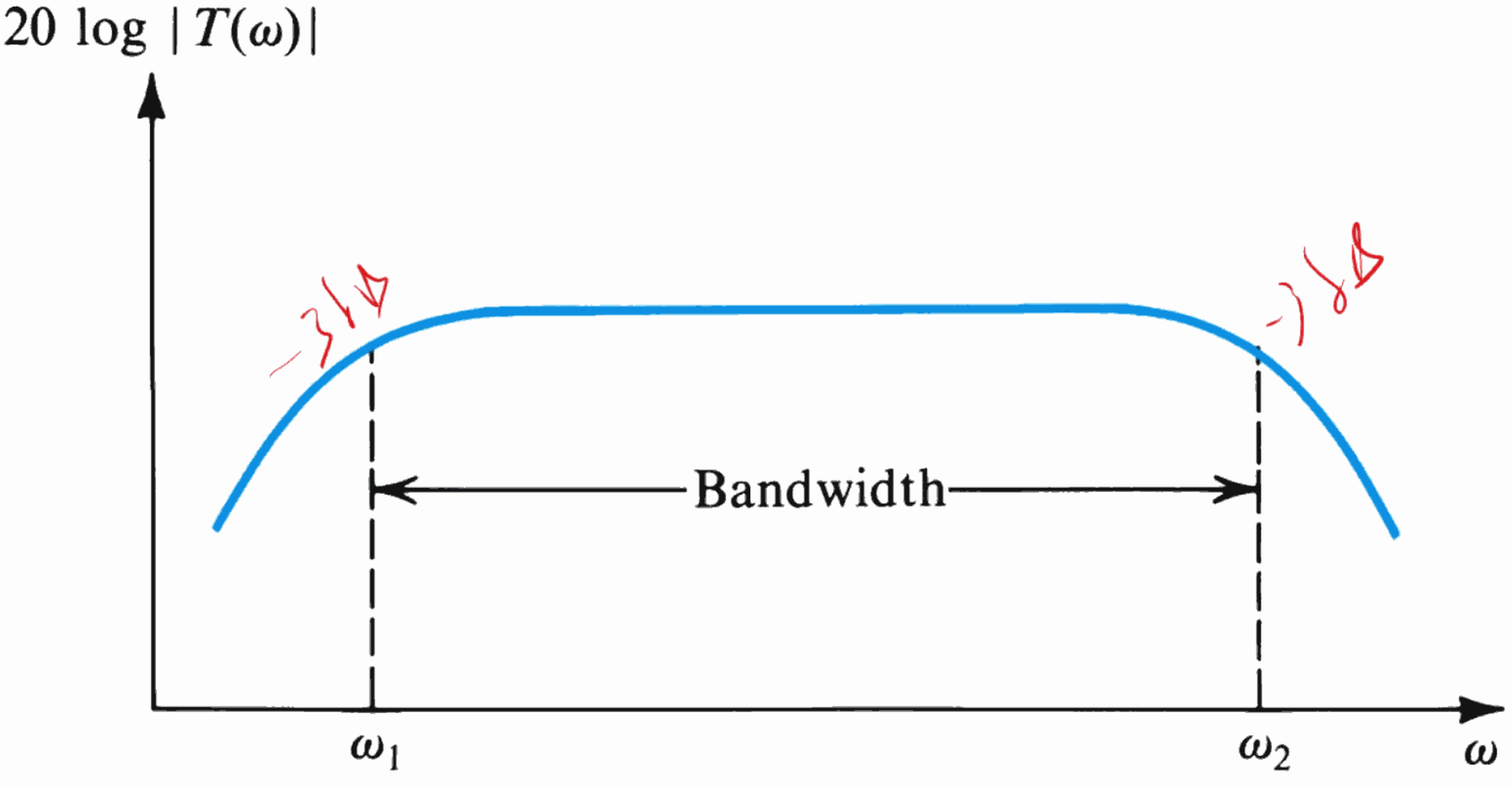
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1.6.4. Single Time-Constant Networks

- **single time–constant (STC) network** – is composed of (or may be reduced to) one reactive component and one resistance.
  - **low pass filter** – attenuates output at high frequencies, allow low to pass
  - **high pass filter** – attenuates output at low frequencies, allow high to pass
- **time constant (τ)** – describes the length of time required for a network transient to settle from step change ($\tau = L / R = RC$)

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1.6.2. Amplifier Bandwidth

- **Q:** What is **bandwidth** of a device?
  - **A:** The range of frequencies over which its magnitude response is constant (within 3dB).
- **Q:** For an amplifier, what is main bandwidth concern?
  - **A:** That the bandwidth extends beyond range of frequencies it is expected to amplify.

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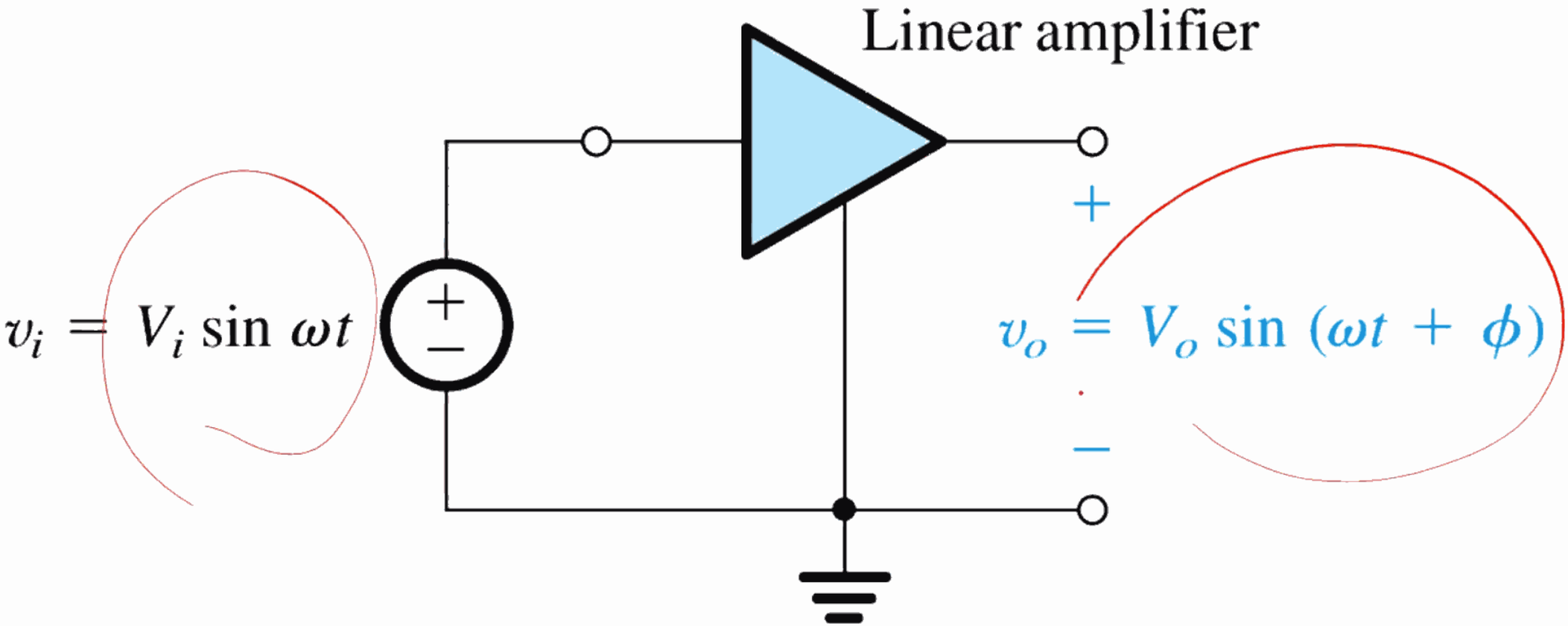
1.6.1. Measuring the Amplifier Frequency Response

- **amplifier transfer function** (**T**) – describes the input-output relationship of an amplifier – or other device – with respect to various parameters, including frequency of input applied.
  - It is a complex value, often defined in terms of magnitude and phase shift.

$$\underbrace{|\mathbf{T}(\omega)| = \frac{V_o}{V_i}}_{\text{magnitude gain}} \quad \text{and} \quad \underbrace{\angle \mathbf{T}(\omega) = \phi}_{\text{phase shift}}$$

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input and output are similar for linear amplifier



1.6.1. Measuring the Amplifier Frequency Response

- **Q:** How does one examine frequency response?
  - **A:** By applying sine-wave input of amplitude V_i and frequency ω.
- **Q:** Why?
  - **A:** Because, although its amplitude and phase may change, its shape and frequency will not.

this characteristic of sine wave applied to linear circuit is unique

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input and output share common terminal

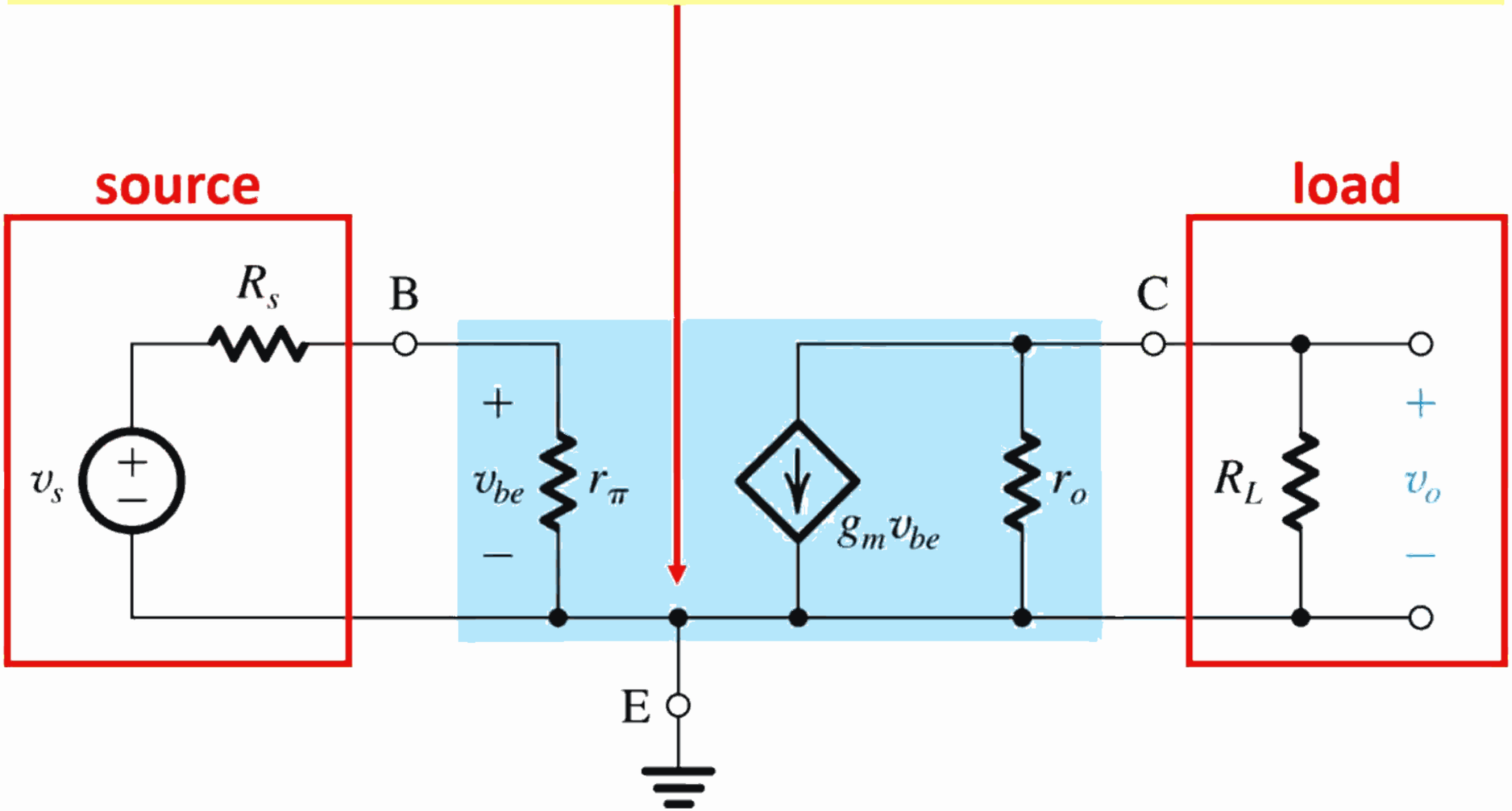


Figure 1.19(b): The BJT connected as an amplifier with the emitter as a common terminal between input and output (called a common-emitter amplifier).

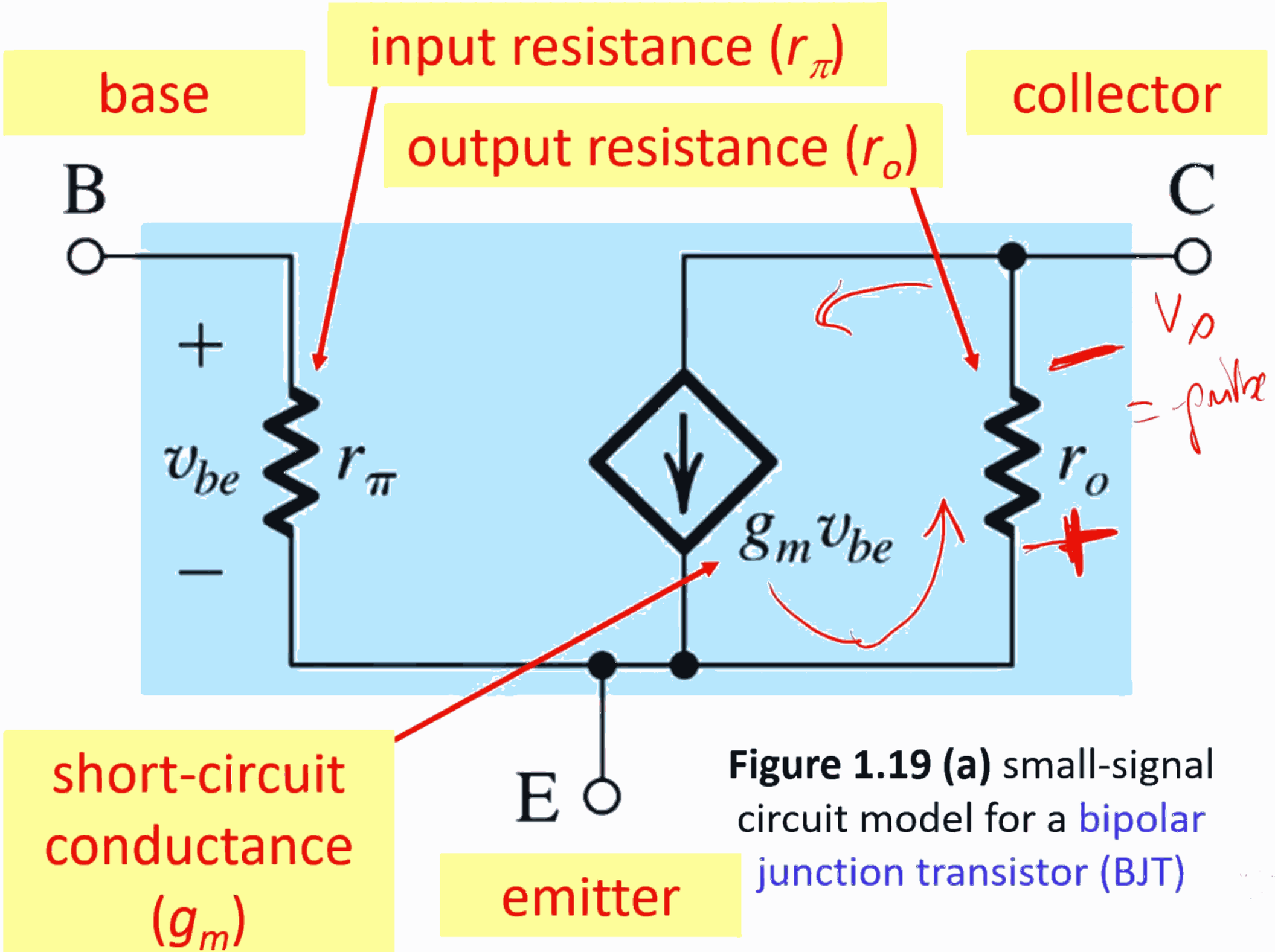
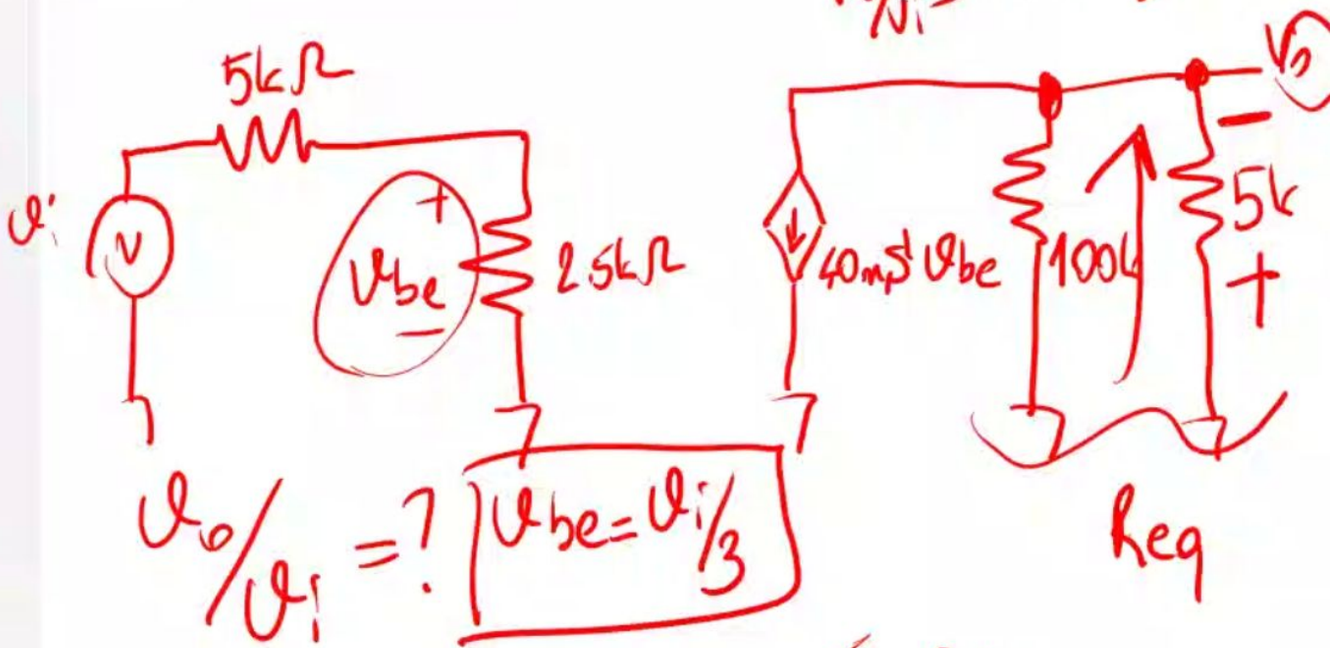


Figure 1.19 (a) small-signal circuit model for a bipolar junction transistor (BJT)

Example 1.4: Common-Emitter Circuit

- **Q(a):** Derive an expression for the voltage gain v_o / v_i of common-emitter circuit with:
  - $R_s = 5kohm$
  - $r_\pi = 2.5kohm$
  - $g_m = 40mA/V$
  - $r_o = 100kohm$
  - $R_L = 5kohm$



$g_m = 40mA/V = 40mS$

$V_o = -R_{eq} \cdot 40mS \cdot v_{be}$

$v_o = -500 \cdot 40mS \cdot \frac{v_i}{3}$

$v_o / v_i = -63.5 \ [V/V]$

$v_o = \frac{40mS \cdot v_i}{3} \cdot \frac{500}{105}$

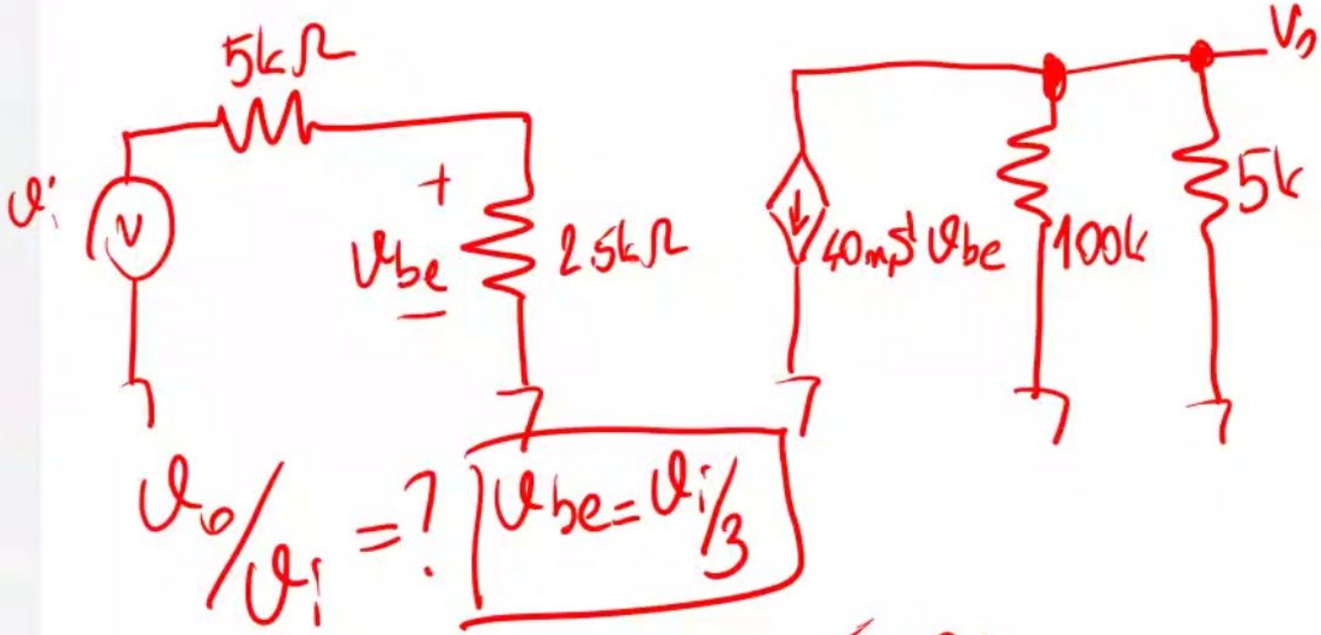
$V_o / V_i = \frac{20.000}{315} = 63,5 \ [V/$

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Example 1.4: Common-Emitter Circuit

$g_m = 40mA/V = 40mS$

- **Q(a):** Derive an expression for the voltage gain v_o / v_i of common-emitter circuit with:
  - $R_s = 5kohm$
  - $r_\pi = 2.5kohm$
  - $g_m = 40mA/V$
  - $r_o = 100kohm$
  - $R_L = 5kohm$

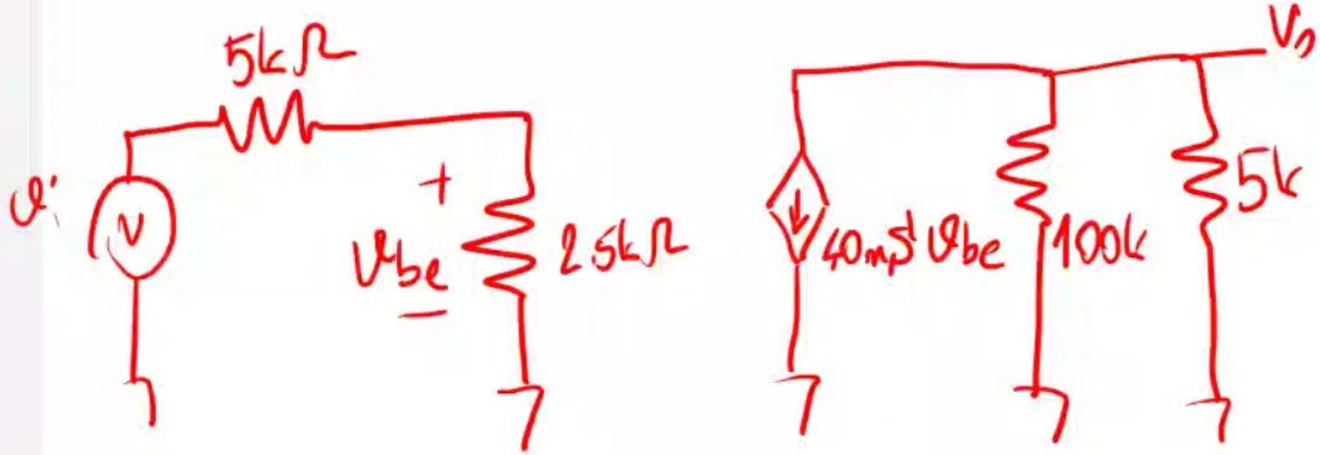


$$v_b = \frac{40mS \cdot v_i}{3} \cdot \frac{500}{105} \quad V_o/V_i = \frac{20.000}{315} = 63,5$$

Example 1.4: Common-Emitter Circuit

$g_m = 40mA/V = 40mS$

- **Q(a):** Derive an expression for the voltage gain v_o / v_i of common-emitter circuit with:
  - $R_s = 5kohm$
  - $r_\pi = 2.5kohm$
  - $g_m = 40mA/V$
  - $r_o = 100kohm$
  - $R_L = 5kohm$



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Example 1.4: Common-Emitter Circuit

- **Q(a):** Derive an expression for the voltage gain v_o / v_i of common-emitter circuit with:
  - $R_s = 5 kohm$
  - $r_\pi = 2.5 kohm$
  - $g_m = 40 mA/V$
  - $r_o = 100 kohm$
  - $R_L = 5 kohm$

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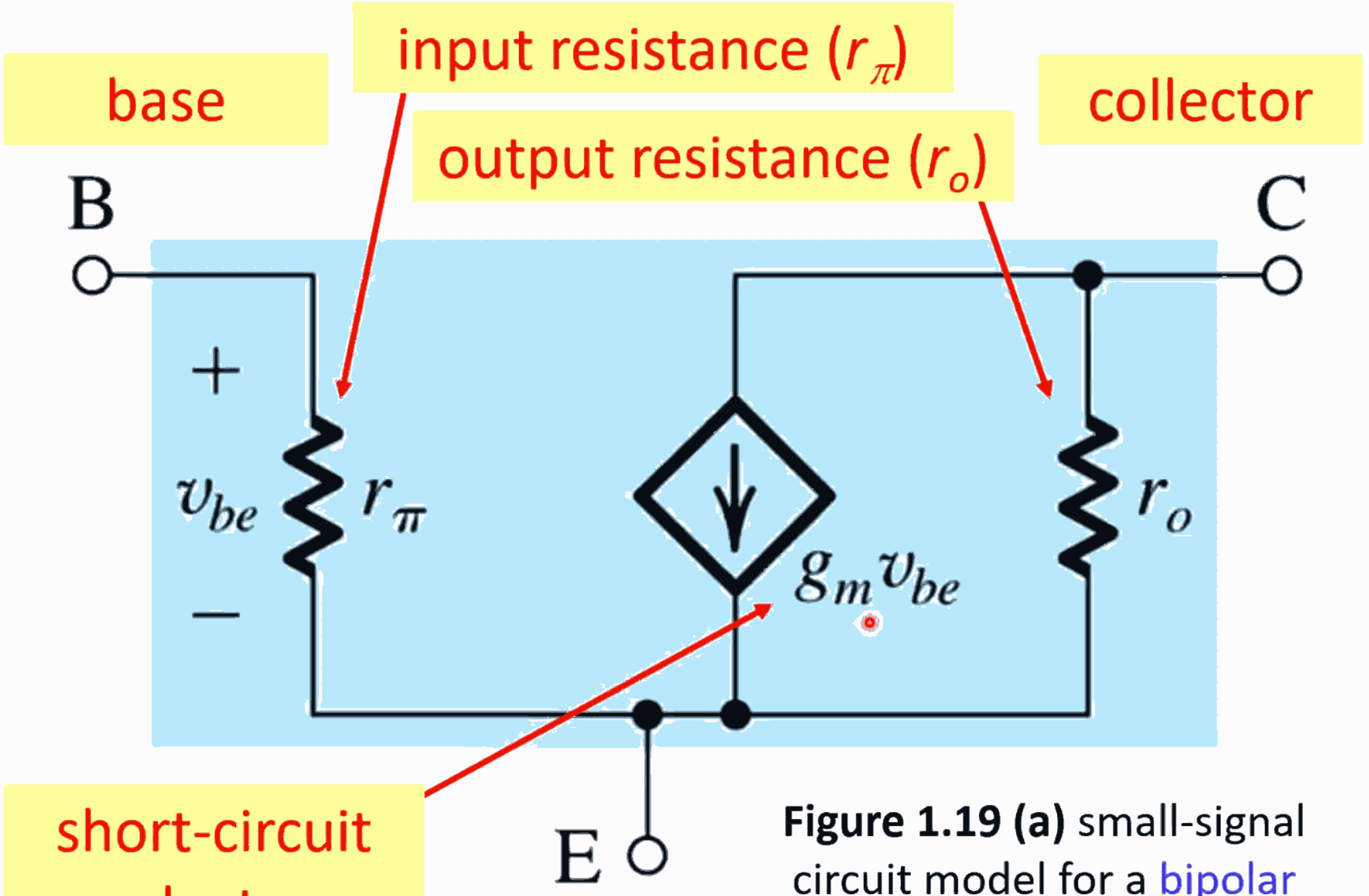
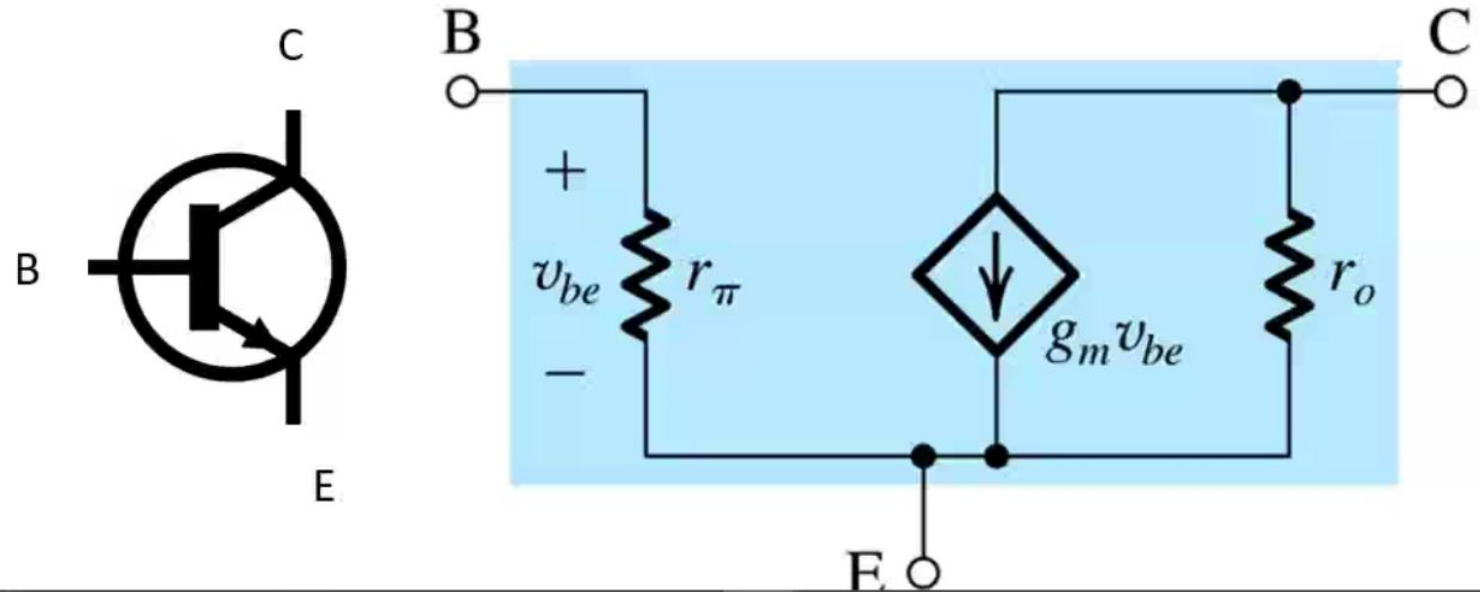


Figure 1.19 (a) small-signal circuit model for a bipolar junction transistor (BJT)

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Example 1.4: Common-Emitter Circuit

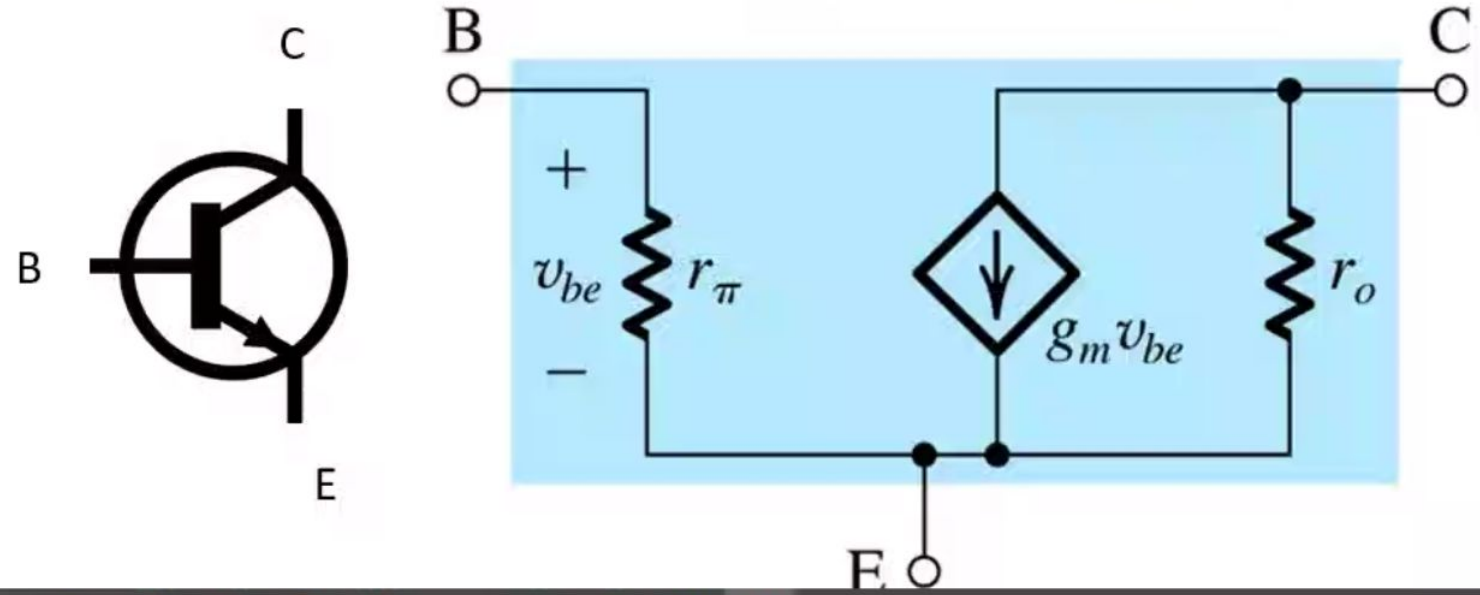
- Examine the **bipolar junction transistor** (BJT).
  - three-terminal device
  - when powered up with dc source and operated with small signals, may be modeled by linear circuit below.



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Example 1.4: Common-Emitter Circuit

- Examine the **bipolar junction transistor** (BJT).
  - three-terminal device
  - when powered up with dc source and operated with small signals, may be modeled by linear circuit below.



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1.5.6. Unilateral Models

- **unilateral model** – is one in which signal flows only from input to output (not reverse)
  - However, most practical amplifiers will exhibit some reverse transmission…

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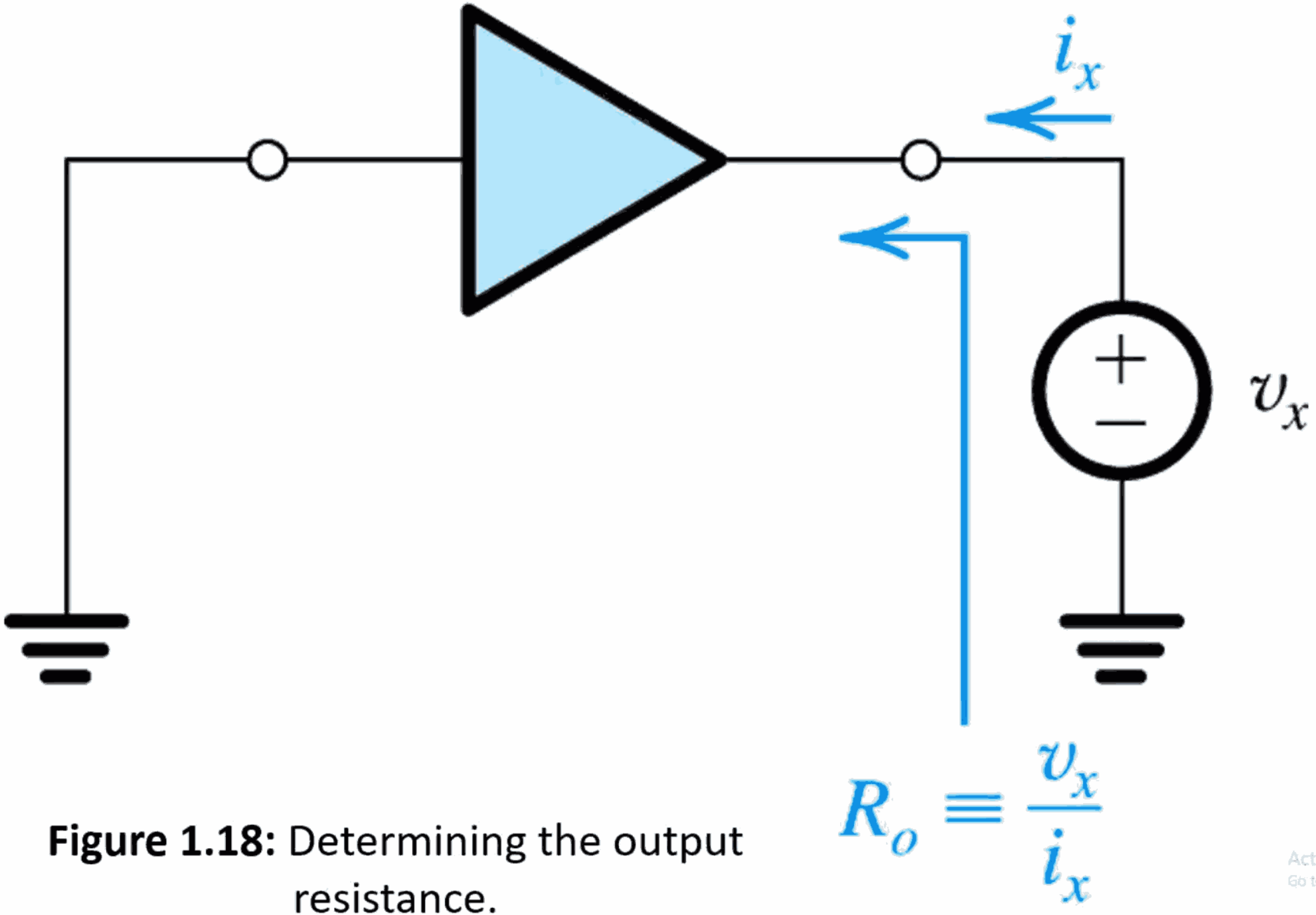


Figure 1.18: Determining the output resistance.

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1.5.5. Determining R_i and R_o

- **Q:** How can one calculate input resistance from terminal behavior?
  - **A:** Observe v_i and i_i, calculate via $R_i = v_i / i_i$
- **Q:** How can one calculate output resistance from terminal behavior?
  - **A:**
    - Remove source voltage (such that $v_i = i_i = 0$)
    - Apply voltage to output (v_x)
    - Measure negative output current ($-i_o$)
    - Calculate via $R_o = -v_x / i_o$

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1.5.4. Relationship Between Four Amp Models

- **interchangeability** – although these four types exist, any of the four may be used to model any amplifier
  - they are related through A_{vo} (open circuit gain)

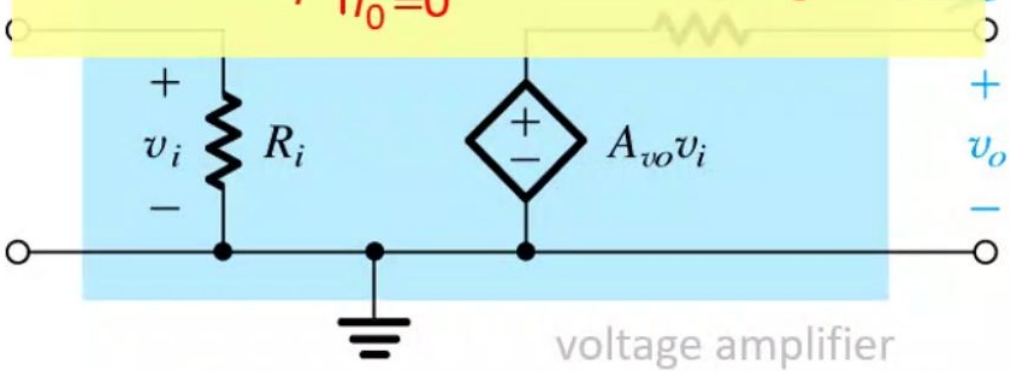
$$A_{vo} = \overbrace{A_{is}\left(\frac{R_o}{R_i}\right)}^{\text{current to voltage amplifier}} = \overbrace{G_m R_o}^{\text{transcond. to voltage amplifier}} = \overbrace{\frac{R_m}{R_i}}^{\text{transres. to voltage amplifier}}$$

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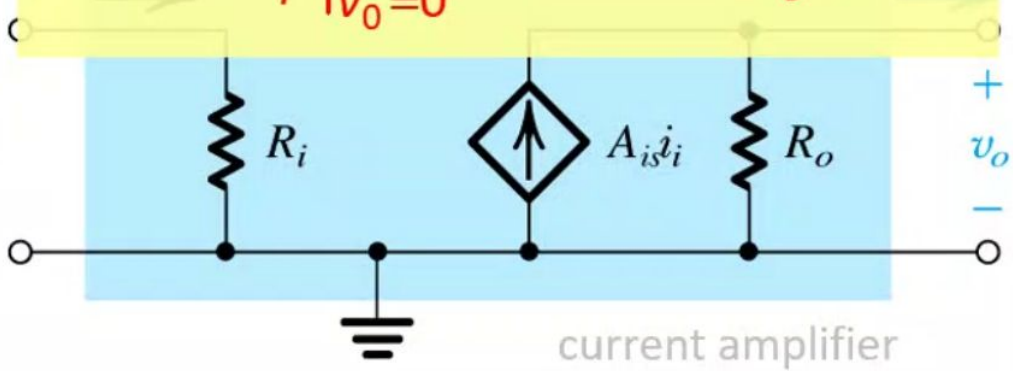
1.5.3. Other Amplifier Types

$$A_{v0} = \left.\frac{v_0}{v_i}\right|_{i_0=0} \quad \text{with} \quad R_i = \infty, \; R_o = 0$$



voltage amplifier

$$A_{v0} = \left.\frac{i_0}{i_i}\right|_{v_0=0} \quad \text{with} \quad R_i = 0, \; R_o = \infty$$



current amplifier

transconductance amplifier



$$G_m = \left.\frac{i_0}{v_i}\right|_{v_0=0} \quad \text{with} \quad R_i = \infty, \; R_o = \infty$$

transresistance amplifier



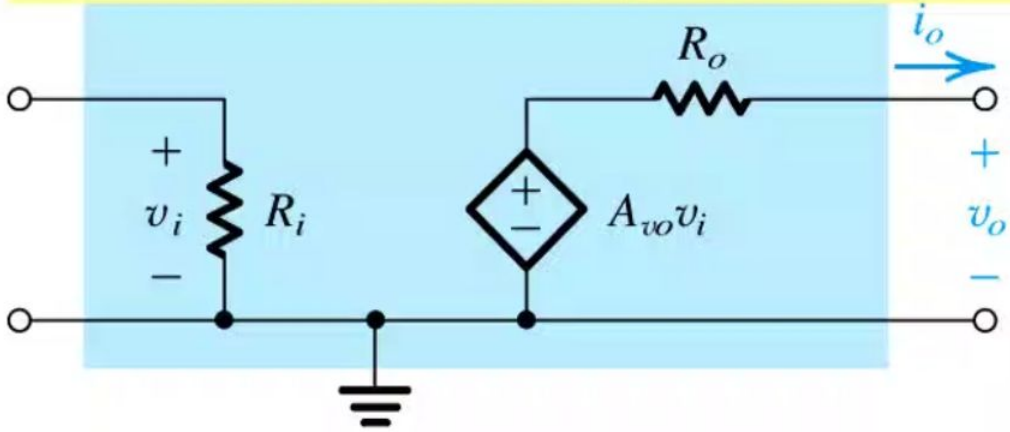
$$R_m = \left.\frac{v_0}{i_i}\right|_{i_0=0} \quad \text{with} \quad R_i = 0, \; R_o = 0$$

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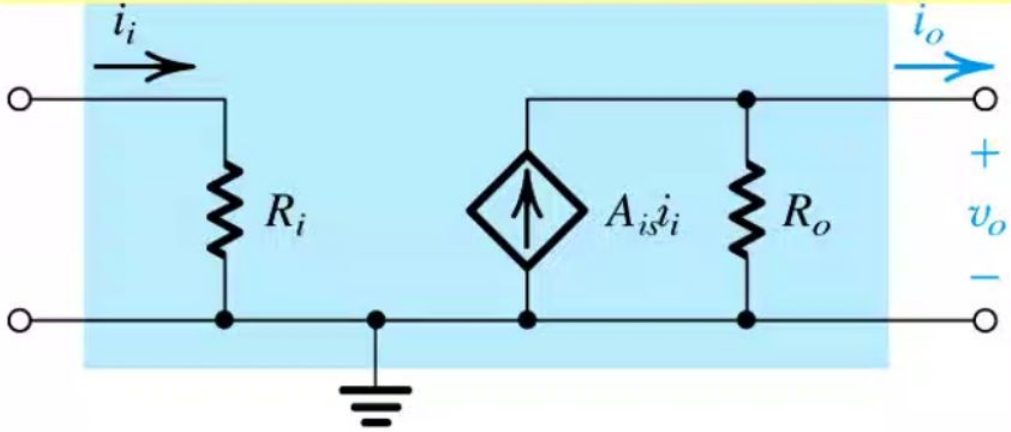
1.5.3. Other Amplifier Types

$A_{is} \cdot \frac{1}{1} = \frac{A_{vo} \cdot \frac{1}{1}}{R_s}$

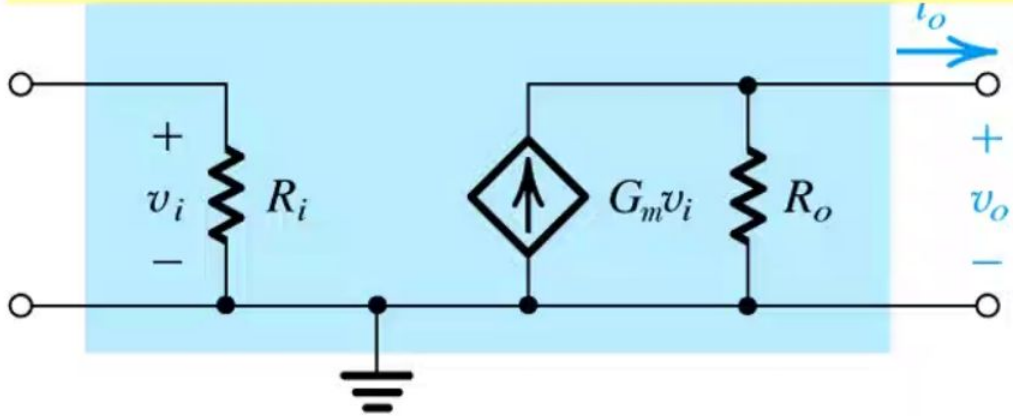
voltage amplifier



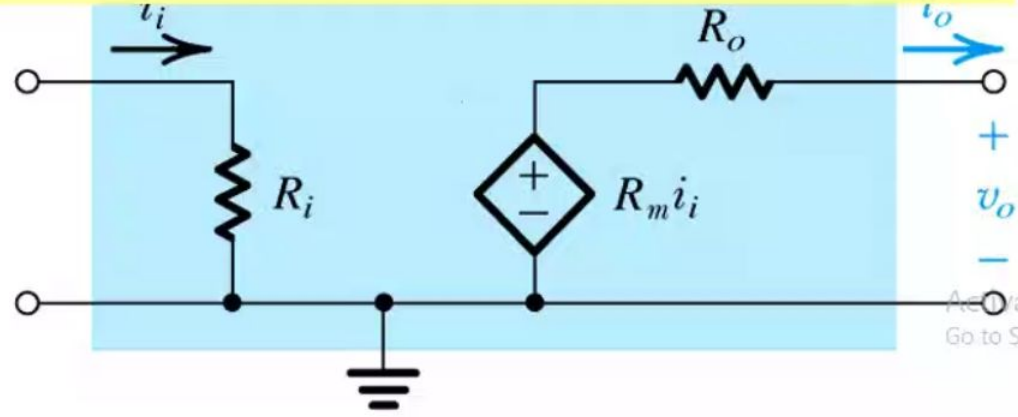
current amplifier



transconductance amp.



transresistance amp.

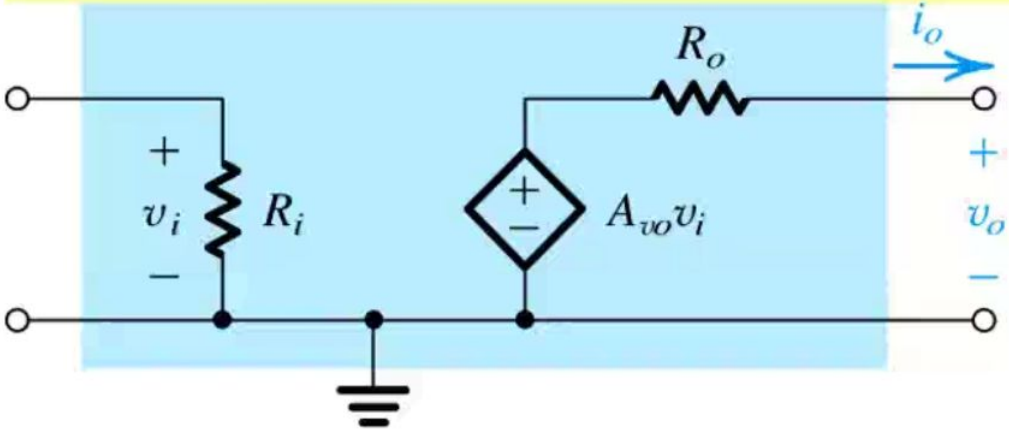


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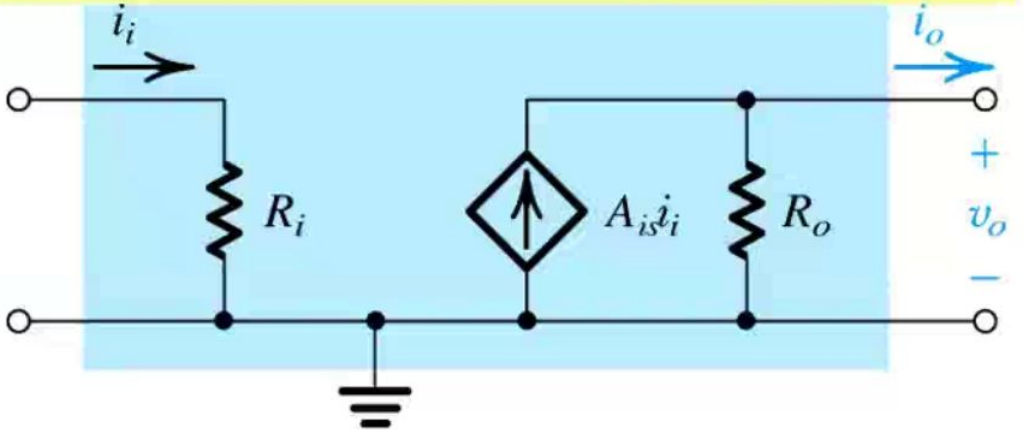
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1.5.3. **Other Amplifier Types**

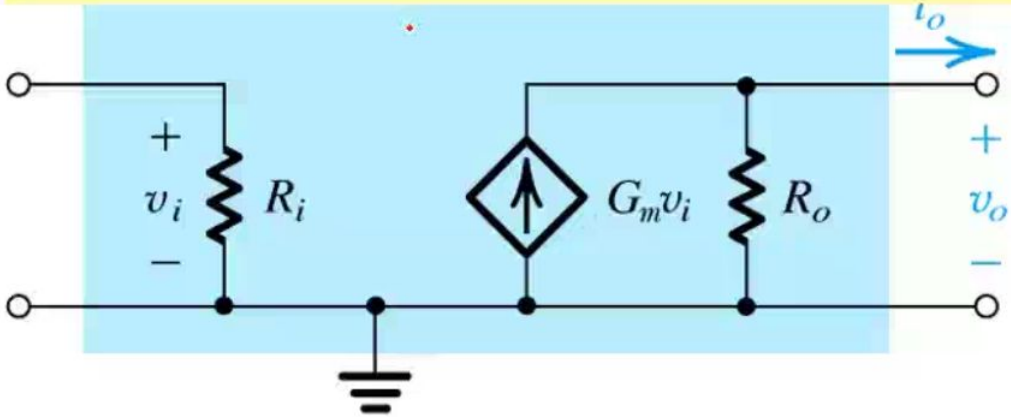
voltage amplifier



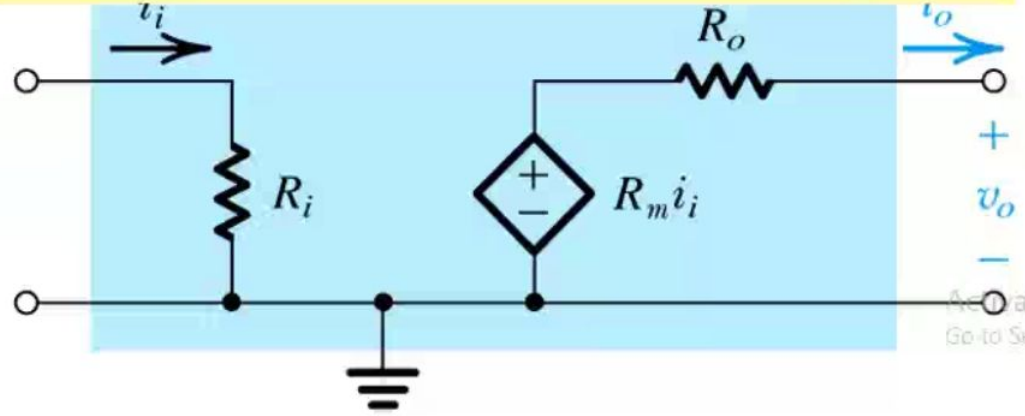
current amplifier



transconductance amp.



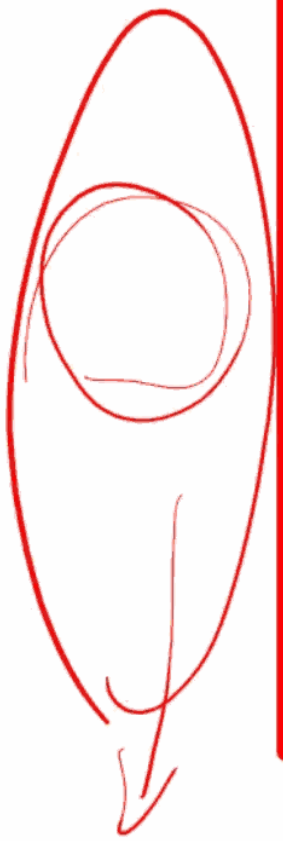
transresistance amp.

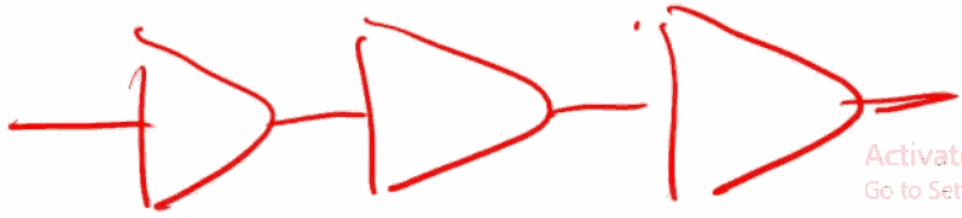


aggregate amplifier

with gain

$$A_v = \frac{v_L}{v_s - i_i R_s}$$





Example 1.3: Cascaded Amplifier Configurations

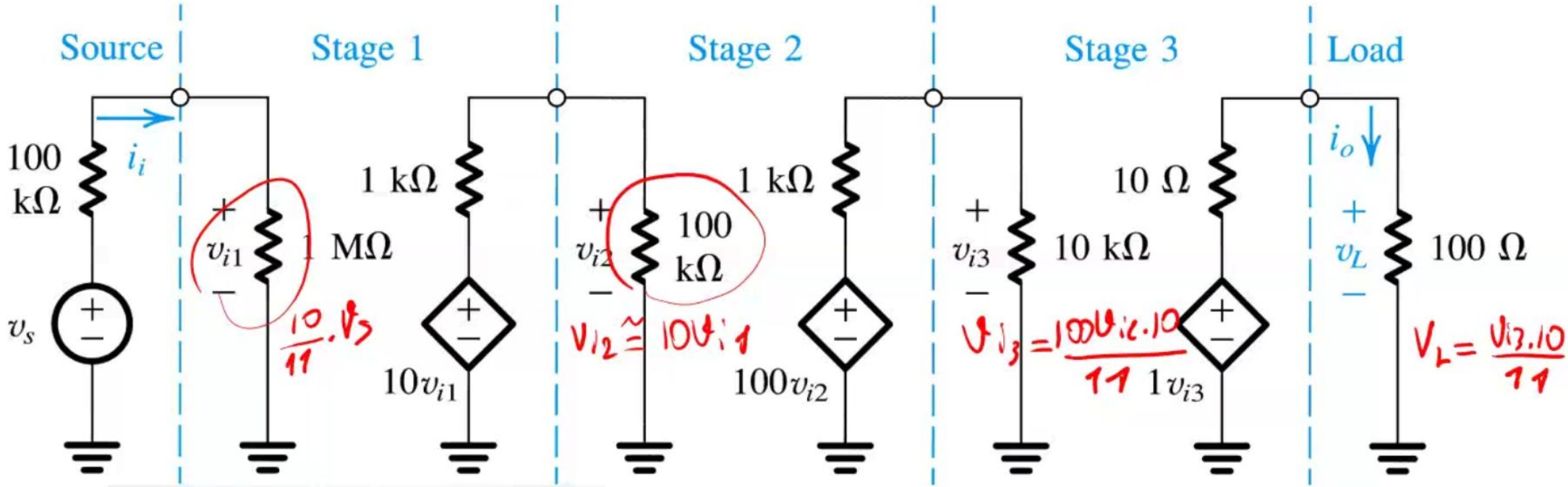


Figure 1.17: Three-stage amplifier for Example 1.3.

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aggregate amplifier

with gain

$$A_v = \frac{v_L}{v_s - i_i R_s}$$

Example 1.3: Cascaded Amplifier Configurations

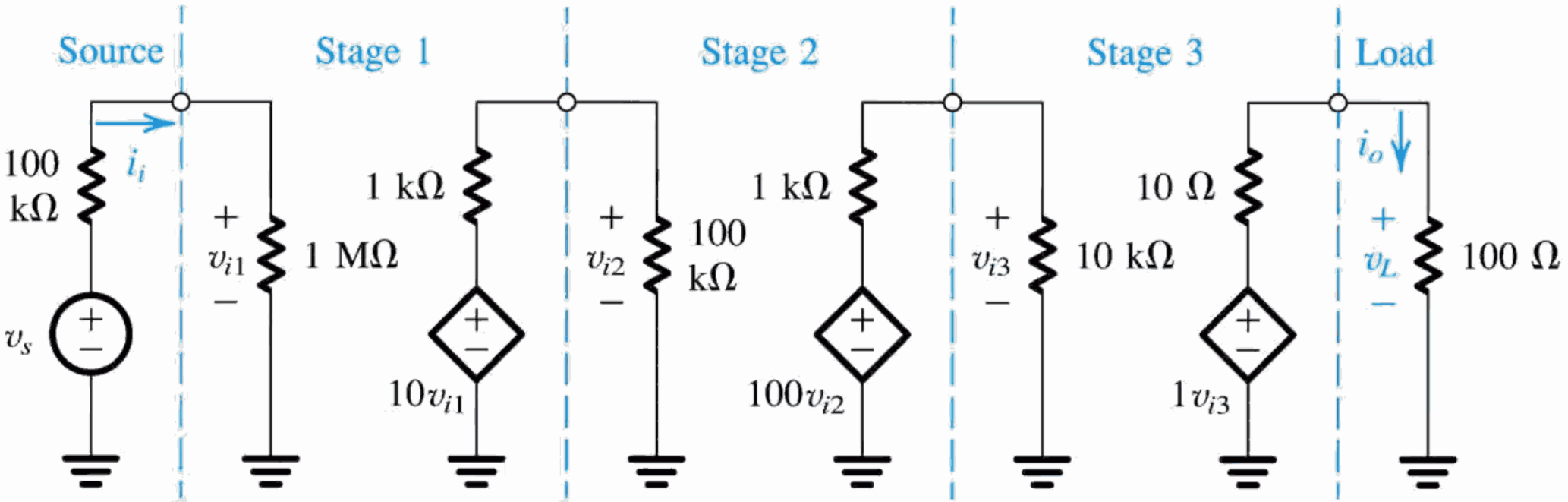


Figure 1.17: Three-stage amplifier for Example 1.3.

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1.5.2. Cascaded Amplifiers

- In real life, an amplifier is not ideal and will not have infinite input impedance or zero output impedance.
- **Cascading** of amplifiers, however, may be used to emphasize desirable characteristics.
  - first amplifier – high R_i, medium R_o
  - last amplifier – medium R_i, low R_o
  - aggregate – high R_i, low R_o

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1.5.1. Voltage Amplifiers

- **ideal amplifier model** – is function of v_s and A_{vo} only!!
  - It is assumed that $R_o << R_L$...
  - It is assumed that $R_i << R_s$...

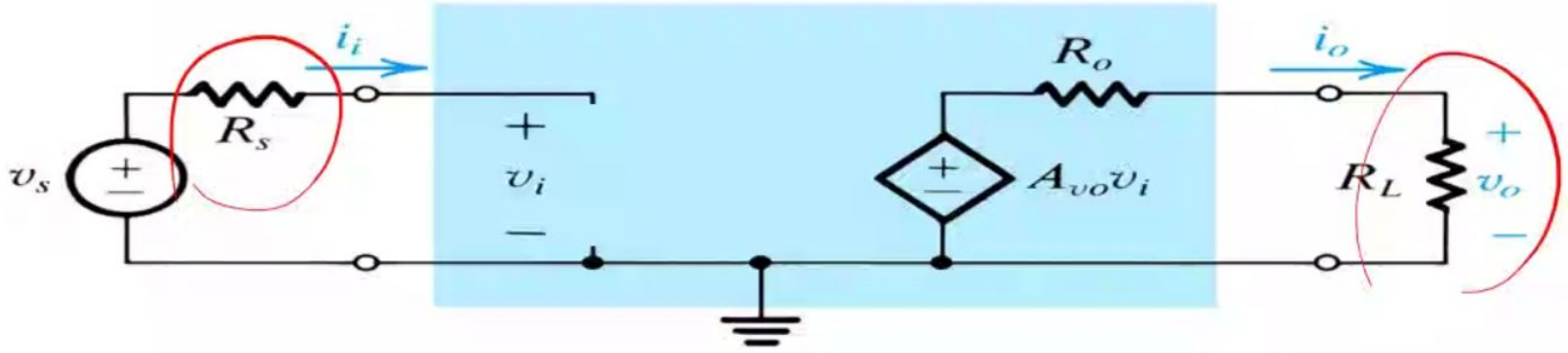
$$v_o = A_{vo} v_s \underbrace{\frac{R_i}{R_i + R_s} \frac{R_L}{R_L + R_o}}_{\text{non-ideal model}} = \underbrace{A_{vo} v_s}_{\text{ideal model}}$$

key characteristics of ideal voltage amplifier model = source resistance R_S and load resistance R_L have no effect on gain

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1.5.1. Voltage Amplifiers

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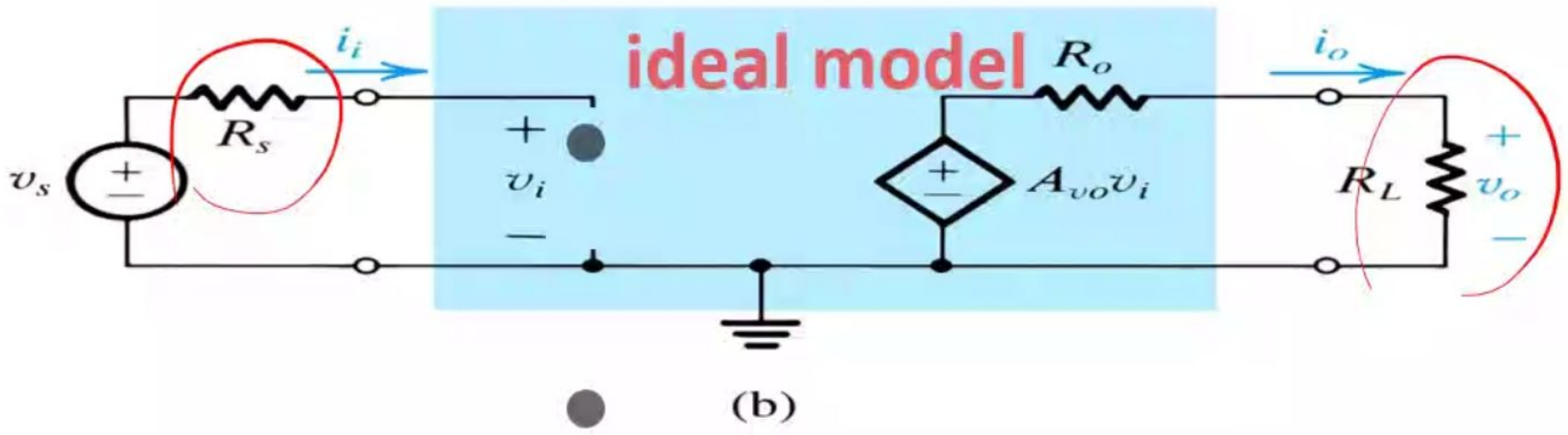


key characteristics of ideal voltage amplifier model = high input impedance, low output impedance

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1.5.1. Voltage Amplifiers

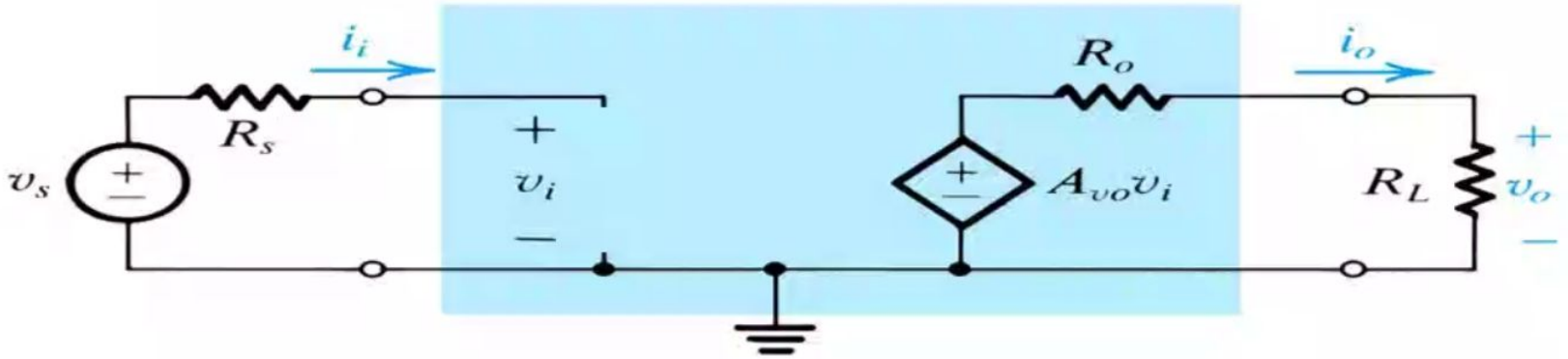
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1.5.1. V
Ampli

- **ideal amplifier**

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1.5.1. Voltage Amplifiers

- **Q:** What is one "problem" with this behavior?
  - **A:** Gain (ratio of v_o and v_s) is not constant, and dependent on input and load resistance.

$$v_o = \left(A_{vo} \underbrace{(v_s)}_{\text{source volt.}} \underbrace{\frac{R_i}{R_i + R_s}}_{\text{source and input resistances}} \right) \underbrace{\frac{R_L}{R_L + R_o}}_{\text{output and load resistances}} = A_{vo} v_s \frac{R_i}{R_i + R_s} \frac{R_L}{R_L + R_o}$$

The **ideal amplifier model** neglects this nonlinearity.

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1.5.1. Voltage Amplifiers

Avo => intrinsic gain

- **Q:** How can one model the amplifier behavior from previous slide?
  - **A:** Model which is function of: v_s, A_{vo}, R_i, R_s, R_o, R_L

$$v_o = \underbrace{\left(A_{vo} \underbrace{(v_s)}_{\text{source volt.}} \underbrace{\frac{R_i}{R_i + R_s}}_{\text{source and input resistances}} \right)}_{\text{open-ckt output voltage}} \underbrace{\frac{R_L}{R_L + R_o}}_{\text{output and load resistances}} = A_{vo} v_s \frac{R_i}{R_i + R_s} \frac{R_L}{R_L + R_o}$$

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1.5.1. Voltage Amplifiers

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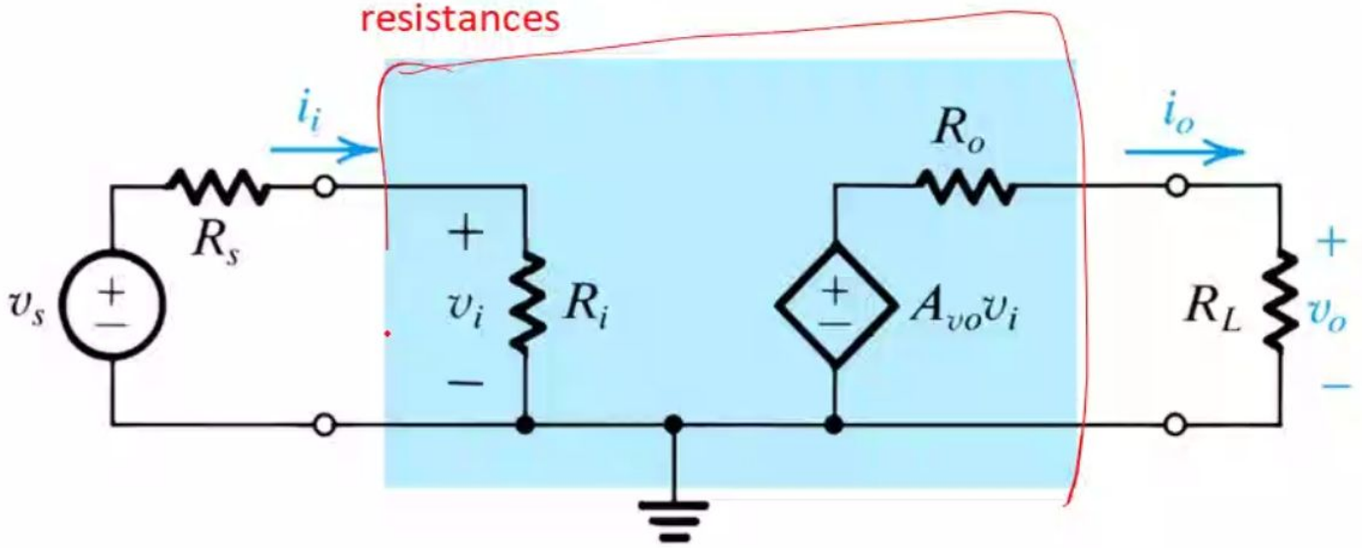
1.5.1. Voltage Amplifiers

model of amplifier input terminals

$$\text{input voltage} = v_i = \underbrace{(v_s)}_{\text{source volt.}} \frac{R_i}{\underbrace{R_i + R_s}_{\text{source and input resistances}}}$$

model of amplifier output terminals

$$\text{output voltage} = v_o = \underbrace{(A_{vo} v_i)}_{\text{open-ckt output voltage}} \frac{R_L}{\underbrace{R_L + R_o}_{\text{output and load resistances}}}$$



(b)

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